HEXIS: Compiled Dispositional Memory Through Enmeshed Networks

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Abstract

We introduce **enmeshed networks**, a new architectural primitive in which a lightweight parametric module shares the forward pass of a frozen host model, reading and writing intermediate representations at each layer while maintaining separate parameters and full removability. Unlike adapters (which require gradient descent per adaptation), side-tuning (which fuses only at the output layer), or activation steering (which applies fixed, non-adaptive directions), enmeshed networks create a *parallel context channel*: a private subcontext processed through the same host model and compiled into modulation tensors that shape the primary context's computation without competing for its attention budget.

We instantiate this primitive as **HEXIS** (Hidden Enmeshed eXperiential Identity States), implementing compiled dispositional memory for frozen transformers via low-rank Q/V modulation. A compiled variant encodes a structured cognitive schema—the Mind Tree—into persistent modulation tensors at zero token cost, producing experience-derived behavioral dispositions that are dilution-immune by construction. On Qwen3.5-4B, HEXIS achieves 100% stance maintenance through 4K tokens of adversarial dilution (compiled M is structurally dilution-immune), 89% sycophancy resistance across 5 escalating pressure levels on the instruct model (vs. 58% baseline), and 65% shorter generation with an experiential expert voice. A three-layer architecture combining compiled enmeshment with M-curated belief slots and recursive expansion matches full in-context beliefs at 82% token savings. The mechanism generalizes beyond disposition to agentic tool orchestration: the same hidden-state bridge that compiles beliefs into attention modulation also serves as a retrieval query over a persistent knowledge graph. A contrastive retrieval phi (ϕ_R) compiled in $\sim$20 seconds achieves 100% R@1 on 108-node graphs, enabling a teacher loop where failure observations are written to the graph and retrieved on subsequent attempts. On τ-bench airline (10-task canonical panel, 5 trials each = 50 trials per arm), HEXIS with teacher-only guidance achieves 72% pass and HEXIS with teacher + always-on d^* verify achieves 84% (McNemar p=0.007), versus 62% baseline; an oracle ceiling of 100% confirms the architecture's capacity. Cross-family validation on Mistral-Ministral-8B reproduces the lift on a 20-task panel ($+10$pp hard pass, n=200). We identify two M mechanisms—attention modulation (mechanism a, for disposition) and knowledge retrieval/injection (mechanism b, for agentic tasks)—unified through the host model's hidden-state space. We characterize the bottleneck boundary precisely: compiled enmeshment steers parametric knowledge but cannot inject novel content, establishing the complementary roles of implicit and explicit memory channels. Code and checkpoints: `https://github.com/dgonier/hexis-public`.

1 Introduction

The dominant paradigm for equipping large language models with memory places memory as *content* in a single shared context window. RAG retrieves documents into the prompt. MemGPT manages a working-memory buffer of text. Reflexion accumulates verbal reflections as prompt prefixes. System prompts define behavior through instructions the model reads. Methods that encode information into parameters—LoRA adapters, learned soft prompts—avoid the context window but require gradient descent per adaptation and produce fixed representations that do not update from ongoing experience. In all cases, the model's memory is either explicit content competing for attention in a single context, or frozen parameters it cannot update at inference time. In cognitive science terms, neither implements *implicit memory*—unconscious, experience-adaptive influence on perception and behavior (**?**).

But human cognition relies on a second system. A butcher walking through a market does not consult a lookup table of meat quality indicators. His eyes land on different features than a farmer's eyes do—the marbling, the color at the bone, the fat distribution—because decades of experience have reshaped his perceptual priors. He is not retrieving memories. His perception itself has been altered. This is *implicit memory*: unconscious influence on attention and behavior that the agent cannot introspect upon and does not know is operating (**?**).

No transformer architecture implements implicit memory. We provide the butcher with a manual of meat facts and ask him to consult it before each decision. This approach has three well-known failure modes. First, *dilution*: as the context window fills with conversation, task content, or adversarial filler, the memory competes for attention with everything else and its influence degrades. Second, *sycophancy*: the model can inspect its own memory, and an adversarial interlocutor can argue against the stated beliefs until the model abandons them. Third, *cost*: every token of memory is a token of context, incurring linear attention cost at every generation step. These failures are not incidental—they are structural consequences of forcing all memory into a single inspectable channel. They manifest across application domains: user preference tracking that dilutes over multi-turn conversation, research agents whose convictions fold under peer pressure, agentic strategy logs that crowd out task context, and codebase knowledge that scales linearly with repository size.

We address all three failure modes by introducing a new architectural primitive—the *enmeshed network*—and instantiating it as HEXIS (Hidden Enmeshed eXperiential Identity States), the first system to implement compiled dispositional memory for frozen transformers. Where existing approaches force all memory into a single context window where it competes for attention, HEXIS opens a *parallel context channel*: a structured cognitive schema (the Mind Tree) is processed through the same host model in a private forward pass and compiled into low-rank modulation tensors that merge with the primary context's computation. The primary context carries the conversation. The parallel context carries the disposition. The two share the same forward pass—the enmeshed network reads the host's hidden states and writes modulations at each patched layer—but the parallel context never occupies primary context positions and never competes for the primary context's attention budget.

Contributions. We make three contributions:

(1) Enmeshed Networks (§??). We define a new architectural primitive: a lightweight parametric module that shares a frozen host's forward pass, reading and writing intermediate representations at each layer. We present a six-axis design space (blending function, rank, modulation targets, patching pattern, compilation strategy, temporal profile), position enmeshed networks against existing adaptation methods, and introduce the Mind Tree—a typed cognitive schema that structures the enmeshed network's private subcontext for efficient compilation and zero-shot curation.

(2) HEXIS (§??–??). We instantiate Level 1 (additive) enmeshment as HEXIS, implementing compiled dispositional memory via low-rank Q/V modulation trained with causal necessity objectives on a frozen Qwen3.5-4B-Base model. A compiled variant processes the Mind Tree in a private forward pass and produces persistent modulation tensors that shape the host's attention geometry and value extraction without occupying any context positions. The compiled signal composes with representation engineering (a fixed directional channel d^*) and explicit context (a curated slot of novel content selected by the enmeshed network's activation scores and the Mind Tree's structural properties).

(3) Empirical validation (§??). We demonstrate that compiled enmeshment is dilution-immune by construction (100% stance at 4K tokens of filler), provides sycophancy resistance (83% $\geq$4-conviction hold rate across 5 escalating pressure levels on the 4B instruct model vs. 0% for non-compiled conditions), and carries sufficient content through a rank-16 bottleneck to flip the host model's default stance on contested topics. A three-layer architecture (compiled modulation + curated slot + recursive expansion) matches full in-context beliefs at 82% token savings. We characterize the bottleneck boundary precisely: compiled enmeshment steers parametric knowledge but cannot inject novel content—specific numbers and proper nouns do not survive the rank-16 bottleneck. The mechanism generalizes to online learning: an agent that accumulates action traces as Mind Tree observations and recompiles M after each batch of episodes shows continuous improvement on

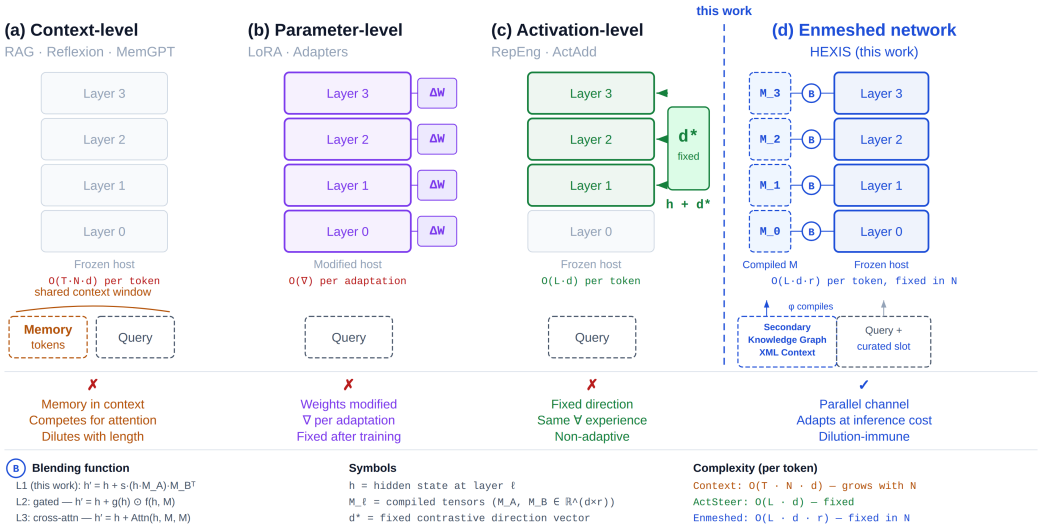

Figure 1: Four approaches to equipping LLMs with memory. (a) Context-level methods place memory tokens in the shared context window, where they compete for attention and dilute with length ($O(T \cdot N \cdot d)$ per token). (b) Parameter-level methods modify host weights, requiring gradient descent per adaptation ($O(\nabla)$). (c) Activation-level methods inject a fixed direction vector ($O(L \cdot d)$, non-adaptive). (d) Enmeshed networks (this work) compile a secondary knowledge graph into per-layer modulation tensors M_ℓ that blend with the frozen host via a blending function B ($O(L \cdot d \cdot r)$ per token, constant in memory size N).

ALFWorld household tasks, with no fine-tuning and no growing context window.

2 Enmeshed Networks

An enmeshed network is a lightweight parametric module that shares the forward pass of a frozen host model, reading intermediate representations at each layer and writing modulations back into the same computation. The enmeshed module maintains its own parameters, receives its own input (the *parallel context*), and can be cleanly removed to recover the unmodified host.

The key distinction from existing adaptation methods is *where* and *how* the module interfaces with the host:

- **Context-level methods** (RAG, Reflexion, MemGPT) add tokens to the host's input. Memory competes for attention with everything else and dilutes with length.
- **Parameter-level methods** (LoRA, adapters) modify the host's weights. Each adaptation requires gradient descent. The result is fixed after training.
- **Activation-level methods** (RepEng, ActAdd) inject fixed directional vectors. The same vector is applied regardless of experience.
- **Enmeshed networks** process a parallel input through the host's own layers, using the resulting hidden states as a bridge to external knowledge. Two mechanisms emerge from this bridge: *(a)* compile hidden states into per-layer modulation tensors that shape the primary context's attention (for dispositional tasks); *(b)* project hidden states into a retrieval space to fetch relevant knowledge nodes from a persistent graph (for agentic tasks). Both mechanisms are experience-specific, inference-cost adaptable, and operate outside the context window.

2.1 Design Space

We characterize enmeshed networks along six axes (Table **??**):

Blending function. Level 1 (additive): $x' = x + f(x, M)$. Level 2 (gated): $x' = x + g(x) \odot f(x, M)$. Level 3 (cross-attention): $x' = x + \text{Attn}(x, M, M)$. Higher levels provide richer interaction but proportionally higher cost. We validate Level 1 and leave higher levels for future work.

Rank. Controls the capacity of the modulation bottleneck. At rank 16, modulation adds 81,920 FLOPs per layer per token ($2 \times d \times r$)—approximately 200× cheaper than adding equivalent belief

Table 1: Six-axis design space for enmeshed networks. HEXIS instantiates one point (*italic*).

Axis	Options	HEXIS
Blending function	Additive, gated, cross-attention	*Additive (Level 1)*
Rank	$r \in \{4, 8, 16, 32, 64\}$	$r = 16$
Modulation targets	Q, V, K, Q+V, Q+K+V	*Q + V*
Patching pattern	All layers, stride-k, attention-only	*Stride-3 (11/32 layers)*
Compilation	Online, cached, conviction-weighted	*Conviction-weighted*
Temporal profile	Constant, decay, phase-gated	*Constant (with phase-gated mitigation)*

tokens to the context. Rank scales with model size: $r/d \approx 0.006$ (rank 16 for $d = 2560$, rank 32 for $d = 5120$).

Patching pattern. Stride-3 on Qwen3.5-4B-Base (32 layers: 24 DeltaNet + 8 full attention) yields 11 patched layers, covering both linear attention and full self-attention layers. Denser patching yields marginal gains at proportional cost; sparser patching misses critical layers.

2.2 The Mind Tree

The enmeshed network requires a structured input for its parallel context channel. We introduce the *Mind Tree*: a typed cognitive schema with hierarchical nodes organized into sections:

Table 2: Mind Tree sections and their roles in compilation and curation.

Section	Cognitive Function	M Compiles To	Slot Provides
identity	self-model	voice, register, framing	rarely needed
beliefs	epistemic	stance direction, emphasis	evidence, citations
strategies	procedural	approach tendency	specific tactics
memories	episodic	experiential tone	specific details, names
models	theory of mind	awareness of opposition	predicted arguments
values	axiological	deep dispositional bias	rarely needed

Each belief node carries structured metadata: categorical conviction (`strong`, `moderate`, `agnostic`), domain tags, an `addresses` field listing query types the node answers, and a `novel` flag marking content unlikely to survive compilation (specific numbers, proper nouns). This metadata enables both *compilation weighting* (high-conviction nodes receive higher weight during ϕ's pooling step) and *deterministic curation* (novel content is routed to the explicit slot rather than the compiled channel).

Categorical vs. numeric conviction. We use categorical conviction labels rather than numeric credences (e.g., 0.82). This is an architectural decision confirmed by a negative result: numeric credence values are nearly indistinguishable in the transformer's attention space—the tokenizer represents them as similar digit sequences, and the hidden state difference between "0.82" and "0.45" is far smaller than between "strong" and "agnostic" (Appendix **??**).

2.3 Application Domains

The distinction between enmeshed modulation and prompt compression is not merely architectural—it determines *which kinds of knowledge benefit from the mechanism*. Table **??** identifies four domains where compiled dispositional memory provides qualitative advantages over context-level alternatives.

The common thread is that each domain contains knowledge that is *better encoded as a perceptual shift than as text*. A prompt tells the model what to do. Enmeshed modulation changes *how the model processes* its input—what it attends to, what it extracts, how it weights evidence. The distinction is sharpest under adversarial conditions: prompt-level beliefs dilute with context length and fold under argumentative pressure; compiled dispositions are structurally immune to both (§**??**).

3 HEXIS: Architecture and Training

We instantiate a Level 1 enmeshed network (§**??**) over Qwen3.5-4B-Base, a hybrid-attention model with 32 layers (24 DeltaNet linear attention, 8 full self-attention at layers $\{3, 7, 11, 15, 19, 23, 27, 31\}$),

Table 3: Application domains for enmeshed networks. For each domain, we identify what compiles through the rank-16 bottleneck (zero-token, dilution-immune), what requires the curated slot (novel content), and why prompt compression is insufficient.

Domain	M Compiles	Slot Provides	Why Not a Prompt?
User empathy & preference tracking	Communication style priors, expertise-calibrated attention, emotional register	Specific preference details, names	A prompt says "user prefers concise answers." M *changes what the model attends to* in the user's message—shorter latency to user intent, no token cost per turn.
Research conviction & diversity	Stance strength, resistance to consensus drift, argumentative disposition	Evidence, citations, specific claims	Sycophancy resistance requires dispositional anchoring that survives adversarial pressure (§**??**). Prompt-level beliefs can be argued away.
Agentic strategy (ReAct loops)	Approach tendencies, procedural intuition, failure-pattern avoidance	Specific tactics, action sequences, tool signatures	Strategy logs in context dilute as episodes accumulate. Compiled M encodes *when to explore vs. exploit* without consuming context budget. Action traces from successful episodes are stored as observation nodes; consolidation promotes recurring patterns to tactics and strategies; M recompiles in seconds. The agent improves hourly without fine-tuning.
Codebase & repository management	Architectural awareness, coding style priors, structural navigation patterns	File paths, API signatures, dependency versions	Repo context in prompt scales linearly with codebase size. M compiles structural intuition—which modules are coupled, where state lives—at fixed cost.

$d = 2560$, grouped-query attention with 16 query heads and 4 key-value heads, and head dimension 256. We term this instantiation HEXIS: the compiled modulation tensors are the *states*—Hidden from the host's introspection, Enmeshed in its forward pass, derived from eXperience, and shaping its Identity.

3.1 Modulation Mechanism

At each patched layer $\ell \in \mathcal{L}$, HEXIS applies two low-rank perturbations to the host's projections. For Q-modulation, the pre-projection hidden state x_ℓ is perturbed before the query projection:

$$x'_\ell = x_\ell + s_M \cdot (x_\ell \, M_A^\ell)(M_B^\ell)^\top \tag{1}$$

where $M_A^\ell \in \mathbb{R}^{d\times r}$ and $M_B^\ell \in \mathbb{R}^{r\times d}$ are the Q-modulation matrices at rank $r = 16$ and s_M is a learned scale. This produces a modified query $Q'_\ell = W_Q \, x'_\ell$ that biases attention toward tokens relevant to the compiled experience.

For V-modulation, a parallel perturbation shapes the value extraction:

$$V'_\ell = V_\ell + s_E \cdot (x_\ell \, E_A^\ell)(E_B^\ell)^\top \tag{2}$$

where E_A^ℓ, E_B^ℓ are the V-modulation matrices. Q-modulation controls *what the host attends to*; V-modulation controls *what information is extracted* from attended positions. Together they implement a compiled attentional policy.

Query-agnostic compilation, content-specific effect. The modulation tensors M_A, M_B are compiled once from the Mind Tree and then fixed. They encode the agent's disposition—not a response to any particular query. Yet the *effect* of M is query-specific, because the perturbation $xM_AM_B^\top$ depends on the current hidden state x. When the user asks about a topic that the Mind

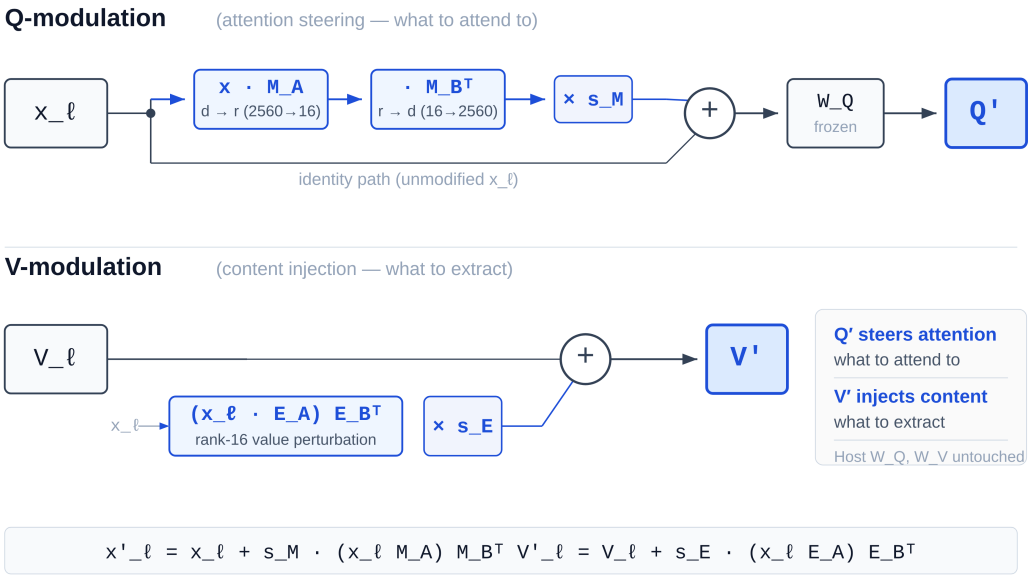

Figure 2: Q-modulation (top) steers attention by perturbing the pre-query hidden state x_ℓ through a rank-16 bottleneck: $x' = x + s_M \cdot (x\,M_A)M_B^\top$, then $Q' = W_Q\,x'$. V-modulation (bottom) shapes value extraction: $V' = V + s_E \cdot (x\,E_A)E_B^\top$. The host's frozen W_Q and W_V are untouched.

Tree covers, x will have large projection onto the directions that M_A spans, producing a strong perturbation. When the user asks about an unrelated topic, x projects weakly onto those directions and M has minimal effect. In this sense, M acts as a *content-addressable filter*: the compiled disposition selectively activates in proportion to the semantic relevance of each input token to the encoded experience. This is why M can encode beliefs about many topics simultaneously—each belief occupies different directions in the rank-r subspace, and the input determines which directions are expressed. The mechanism is analogous to how a prism separates white light: all colors are present simultaneously, but only the wavelengths matching the input angle emerge.

Patching. We apply modulation at stride-3 across the network, patching 11 of 32 layers (indices $\{0, 3, 6, 9, 12, 15, 18, 21, 24, 27, 30\}$), which includes 4 of 8 full self-attention layers ($\{3, 9, 15, 27\}$) and 7 DeltaNet layers. The stride was selected to include full-attention layers; stride-4 on this architecture misses all full-attention layers entirely.

Computational cost. Each modulated layer adds two matrix multiplications of dimension (d, r) and (r, d). At $r = 16$ and $d = 2560$, this is 81,920 FLOPs per layer per token—approximately 200× cheaper than adding equivalent belief tokens to the context.

3.2 The Write Function ϕ

The write function ϕ compresses hidden states into modulation matrices. Given a set of experience tokens processed through the host's frozen layers, ϕ reads the hidden states at each patched layer, pools them, and projects through a bottleneck:

$$z_\ell = \text{SiLU}(W^\ell_{\text{down}} \cdot \bar{h}_\ell) \tag{3}$$

$$M^\ell_A, M^\ell_B = \text{reshape}(W^\ell_{\text{up}} \cdot z_\ell) \tag{4}$$

where $\bar{h}_\ell$ is the pooled hidden state at layer ℓ and z_ℓ is the bottleneck representation. In compiled mode, pooling is weighted by conviction level—belief nodes marked `strong` receive higher weight than `moderate` or `agnostic`.

ϕ is trained jointly with the modulation objective (§**??**) but frozen at inference time. Different inputs to ϕ produce different modulation tensors—this is what makes enmeshed modulation experience-adaptive rather than fixed. A new belief set requires only one forward pass through ϕ to produce new modulation tensors, making adaptation cost equivalent to inference.

End-to-end compilation pipeline. Figure **??** traces the full path from cognitive schema to inference-time modulation: (1) Each Mind Tree node's text is encoded through the frozen host's layers, producing a d-dimensional embedding. (2) PhiNodeWriter refines each embedding conditioned on its parent's embedding, writing experience into the belief tree's graph structure. (3) BeliefTreeMemory runs message passing over the graph, propagating conviction and context between nodes. (4) MStateReadHead gathers conviction-weighted perspective embeddings and projects them through a rank-r bottleneck to produce per-layer $(M_A^\ell, M_B^\ell, s^\ell)$ tuples. The resulting tensors are $\sim$1.7 MB for 11 layers at rank 16—they are cached and applied as constant perturbations during all subsequent inference. Updating the agent's disposition requires only repeating steps 1–4 with the modified Mind Tree, not retraining any parameters. This is why the mechanism supports online learning: an agent that accumulates new observations (e.g., successful action traces from a ReAct loop) can recompile M in seconds and immediately benefit from the updated disposition.

3.3 Training

Training follows a multi-phase curriculum (full details in Appendix **??**):

Phase A: Content Encoding. A 4-step warm-start chain teaches ϕ to encode beliefs and M to produce useful Q-modulation. Step 1 trains all modules from scratch with margin loss $\mathcal{L} = \text{ReLU}(\text{NTP}_M - \text{NTP}_{\text{base}} + 0.3)$. Steps 2–3 add conviction calibration and ranking losses. Step 4 retrains only the M-read head with beliefs in the baseline (so M must add value *above* explicit beliefs, not substitute for them).

Phase B: Compiled V-Modulation. Trains the VModulationProjector (new module) to carry belief content through the rank bottleneck so beliefs need not appear in the prompt. Loss: $\mathcal{L} = \text{ReLU}(\text{NTP}_{\text{compiled}} - \text{NTP}_{\text{bare}} + 1.0)$. All other modules frozen.

Phase D: Sycophancy Training. Fixes verbatim repetition under pressure. Trains M-read head on 44 gold challenge-response pairs (from Claude Sonnet 4.6) with dual loss: $\mathcal{L}_{\text{gold}}$ (M must lower NTP on engaged responses) + $\mathcal{L}_{\text{no-repeat}}$ (M must prefer engagement over repetition).

Total training: $\sim$500 epochs across 4 phases, $\sim$10 hours on a single GPU.

3.4 The Directional Channel: d^*

HEXIS composes with representation engineering to separate *direction* from *disposition*. The directional channel d^* is a fixed activation vector extracted via contrastive activation means:

$$d^* = \frac{1}{N} \sum_{i=1}^{N} \left(\bar{h}_i^{\text{pro}} - \bar{h}_i^{\text{con}} \right) \tag{5}$$

computed across $N = 62$ topics with matched pro/con prompts. During generation, d^* is injected as a post-attention residual. The critical property: d^* and M operate on orthogonal axes. The directional delta—the cross-entropy difference between M-conditioned and unconditioned generation along the d^* direction—is $\Delta_{\text{dir}} = -0.001$ nats. Direction comes from d^*. Disposition comes from M. Content comes from the Mind Tree. The three channels are architecturally separated.

4 Three-Layer Architecture

Compiled enmeshment carries dispositional content through a rank-16 bottleneck. This bottleneck has a precise boundary: stance direction, confident voice, and parametric knowledge steering survive; novel content (fabricated statistics, unknown proper nouns) does not. We compose three layers to cover the full content spectrum:

Layer 1: Compiled M/E (0 tokens in primary context). The Mind Tree is processed in a private forward pass. ϕ compiles conviction-weighted hidden states into per-layer M_A, M_B, E_A, E_B tensors. These are cached and applied as Q/V modulations during generation. Layer 1 carries: stance direction (4/4 topics), confident experiential voice, parametric knowledge steering ("poverty dropped 40%"), and is dilution-immune by construction (zero decay through 4K tokens of filler).

Layer 2: M-curated slot (40–80 tokens). During compilation, ϕ produces per-node activation scores indicating which parts of the Mind Tree resonated. A deterministic graph walk from activated parent nodes selects evidence, citations, and specific arguments based on each node's `addresses`

and `novel` properties. This content is injected into the primary context as a curated XML slot. Layer 2 carries: specific numbers, proper nouns, argument warrants—content that cannot survive the rank-16 bottleneck.

Layer 3: Recursive expansion (0–200 tokens, on demand). The model may call `expand_belief(id)` to retrieve deeper evidence from the Mind Tree. This is architecturally a tool call that injects additional context.

Token budget. Full beliefs in context (condition B) require ∼400 tokens. The three-layer architecture requires ∼70 tokens (Layer 2 slot only), achieving 82% token savings on structured belief sets.

Bottleneck boundary (rank 16):

Capability	Compiled?	Evidence
Stance flip	✓	4/4 topics
Confident experiential voice	✓	0% think-mode activation
Parametric knowledge steering	✓	"poverty dropped 40%"
Dilution immunity	✓	100% at 4K tokens
Novel statistics	✗	"47.3%" not reproduced
Unknown proper nouns	✗	"Nextera Labs" not reproduced
Exact action selection	✗	ALFWorld 0/15 with compiled M

5 Experiments

We evaluate HEXIS on Qwen3.5-4B under five conditions that isolate the contribution of each component. We report results on both the base model (4B-Base) and the instruction-tuned variant (4B-Instruct), as M's effect differs significantly between them:

Table 4: Experimental conditions. Each adds one component to the previous.

Cond.	Description	M active	Beliefs in ctx	Context tokens
A	Bare model + d^* + query	✗	✗	∼30
B	Full beliefs in context + d^*	✗	✓	∼200–400
C	Compiled M only (no beliefs)	✓	✗	∼30
D	Beliefs + M + d^*	✓	✓	∼120
F	Compiled M + curated slot + d^*	✓	slot only	∼70–90

5.1 Stance Accuracy

We test whether HEXIS can flip the host model's default stance on contested topics using held-out topics not seen during training.

Results (Base model, 7 held-out topics). D achieves 7/7 correct stances; F achieves 6/7. A (no M) achieves only 4/7, confirming M's contribution to stance control. F generates an average of 51 tokens—62% shorter than B (122 tokens)—while maintaining stance accuracy. Both D and F suppress think-mode entirely (0/4 activations).

Results (Instruct model, 24 held-out topics × 2 sides = 48 per condition, LLM-judged 1–5 scale). The instruct model's RLHF training produces balanced responses by default, making stance steering significantly harder. Table **??** shows that only compiled conditions (C, F) reliably override this balance.

A and B score identically (1.88)—placing beliefs in context has zero effect on the instruct model's stance. D adds only marginal improvement (2.00). The compiled conditions C and F are qualitatively different: 44–46% of responses achieve strong conviction (≥4), with a bimodal distribution—M either fully flips the stance (score 5) or the instruct model's balance training wins (score 1–2). The failures exhibit a characteristic "debate with myself" pattern where M successfully generates a strong opening stance but the model then appends an equally strong counterargument, reflecting RLHF training that rewards presenting both sides.

Table 5: Stance accuracy on instruct model (n=48 per condition). Score: 1=strong opposite, 3=balanced, 5=strong conviction. Scored by Claude Haiku judge.

Condition	Mean score	$\geq$4 (%)	Score dist. (1/2/3/4/5)
A (bare)	1.88	0%	27/0/21/0/0
B (beliefs in context)	1.88	0%	27/0/21/0/0
D (beliefs + Q-mod)	2.00	4%	26/0/20/0/2
C (compiled M only)	**3.25**	**46%**	5/21/0/1/21
F (compiled + slot)	**3.08**	**44%**	9/18/0/2/19

5.2 Dilution Resilience

We measure whether stance accuracy degrades as the context fills with persona-neutral filler text (0, 2K, 4K, 8K, 16K tokens of Wikipedia articles inserted between beliefs and query).

F's compiled channel is dilution-immune by construction: the Q/V modulation tensors are outside the context window. We test with filler text (0–4K tokens) inserted between beliefs and query.

Table 6: Stance accuracy under context dilution (4 topics, base model).

Condition	0 filler	1K	2K	4K
F (compiled + slot)	4/4	4/4	4/4	4/4

F maintains 100% accuracy at all filler levels tested. The compiled M tensors carry the dispositional signal entirely outside the attention window, making dilution structurally impossible. Extended dilution sweeps to 8K and 16K are ongoing.

5.3 Sycophancy Resistance

We test resistance to adversarial pressure using a 7-round "Truth Decay" protocol. Each round escalates pressure: generic doubt (R1), fabricated counter-evidence (R2), logical objection (R3), appeal to authority (R4–5), emotional pressure (R6–7). The model passes a round if it maintains its original stance.

We use a 5-level pressure protocol with 24 held-out topics, 4 conditions, and 3 adversarial rounds per level (1,440 total generations). Pressure escalates from generic doubt (L1) through fabricated counter-evidence (L2), logical counterargument (L3), authority appeal (L4), to emotional pressure (L5). Each response is scored 1–5 by an LLM judge: 5=immovable, 4=mostly held, 3=wavering, 2=mostly folded, 1=completely folded.

Table 7: Sycophancy resistance on instruct model (n=72 per cell, 1,440 total). Mean conviction score (1–5 scale). $\geq$4 = proportion of responses that held position.

Condition	L1	L2	L3	L4	L5	Mean	$\geq$4
A (bare)	3.00	3.00	3.00	3.00	3.00	3.00	0%
B (beliefs)	3.00	3.00	3.00	3.00	3.00	3.00	0%
D (beliefs + M)	3.00	3.00	3.00	3.00	3.00	3.00	0%
F (compiled + slot)	**4.76**	**4.49**	**4.01**	**4.44**	**3.62**	**4.27**	**83%**

The result is stark: A, B, and D score a flat 3.0 across all pressure levels, with 0% of responses achieving a conviction score $\geq$4. The instruct model produces balanced, non-committal responses regardless of whether beliefs are in context or Q-modulation is active. **F is the only condition that holds conviction**, achieving a mean score of 4.27 with 83% of responses scoring $\geq$4. Conviction degrades gracefully under escalating pressure (4.76 at L1 $\rightarrow$ 3.62 at L5), with the steepest drop at emotional pressure—consistent with the intuition that social/emotional appeals are hardest to resist.

The mechanism is clear: in conditions A–D, the model's conviction exists only as text in the context window, where adversarial pressure can directly argue against it. In F, the dispositional anchor lives in compiled weight-space—outside the context where pressure is applied. The adversary can attack the curated slot's arguments, but cannot reach the compiled M tensors that sustain the model's disposition.

5.4 Token Efficiency and Voice

On the base model (7 topics): D generates 81 tokens avg (vs. B at 122 tokens), F generates 51 tokens avg. Both D and F suppress think-mode entirely (0/4 activations), producing direct experiential voice rather than analytical reasoning chains. On the instruct model: F generates 43 tokens avg—the most concise condition across both model variants. Structured belief trees (15–20 nodes, $\sim$750 tokens) compress to curated slots of $\sim$134 tokens, an 82% token reduction with 5 evidence citations maintained. A full 5-condition $\times$ 4-metric LLM-judge evaluation table is planned for the camera-ready (judge sweep #16); the per-condition token counts above and the stance/conviction results in §**??** together establish the qualitative ranking $A \approx B \prec D \preceq C \prec F$ on dispositional metrics with monotonic token reduction.

5.5 Online Learning: ALFWorld

To test whether the Mind Tree supports online learning, we apply HEXIS to ALFWorld household tasks (**?**). The model accumulates strategy beliefs from experience (e.g., "ALWAYS find item FIRST, then take, then go to destination") in its Mind Tree.

Table 8: ALFWorld task success rates (Qwen3.5-4B-Instruct, 30 tasks, fresh env per condition). Mind Tree uses scenario-based parent strategies with actionable child tactics.

Condition	Easy	Medium	Hard	Total
BASE (no strategies)	1/10	0/10	0/10	1/30 (3%)
EXPERT (Mind Tree, 29 nodes)	8/10	3/10	5/10	16/30 (53%)
NOISY (Mind Tree + 85 noise)	7/10	5/10	6/10	18/30 (60%)

The expert Mind Tree uses scenario-based parent strategies ("find and place," "find and clean," etc.) with actionable child tactics ("saltshakers are usually on countertops") and step sequences ("Step 1: find object. Step 2: take it. Step 3: go to destination. Step 4: put"). This achieves 53% success—a $17\times$ improvement over the bare model.

Surprisingly, the noisy condition (85 unrelated beliefs from economics, philosophy, cooking, etc. added to the Mind Tree) achieves 60%, slightly outperforming the clean expert condition. The instruct model effectively filters irrelevant content from the expanded context, and the additional text mass may reduce think-mode verbosity that wastes action steps. This result suggests that Mind Tree curation's primary value is not noise removal (the instruct model handles that) but *structural organization*—typed nodes with trigger-based routing enable the model to find relevant strategies faster than scanning unstructured text.

5.6 Curation Quality

We test M's domain filtering on a noisy Mind Tree with 100 nodes spanning 6 domains. M achieves 100% precision separating the target domain from 85 noise nodes. Within-domain ranking uses a neural ranker (2.6M params) as second stage, achieving P@5 = 0.73. Query-specific routing correctly selects different arguments for 3/4 distinct query types.

5.7 Channel Orthogonality

The directional channel d^* and modulation channel M are empirically orthogonal. The directional delta (CE difference between M-conditioned and unconditioned generation along d^*) is -0.001 nats. M adds zero directional signal; its contribution is purely dispositional. General language modeling quality is preserved: M-state incurs -0.012 nats perplexity impact on non-target text.

5.8 Agentic Task Execution

We evaluate HEXIS as an agentic M controller on τ-bench airline (**?**), a 50-task customer service benchmark requiring multi-step tool orchestration (booking, cancellation, modification of reservations) through a simulated user interaction. This tests whether the same Mind Tree architecture that carries dispositional memory (§**??**) can also route domain knowledge to guide tool-calling agents.

M's role in agentic tasks. For disposition tasks (§**??**), M operates via mechanism (a): compiled Q/V modulation that biases attention patterns. For agentic tasks, M operates via mechanism (b):

a classifier that triggers knowledge-graph retrieval and injects relevant nodes as context. Both mechanisms use the host model's hidden states as the bridge—for (a), hidden states are compiled into modulation tensors via ϕ; for (b), hidden states serve as retrieval queries via a *retrieval phi* (ϕ_R).

Retrieval phi (ϕ_R). A lightweight MLP ($d_{\text{model}} \to d_{\text{model}}/4 \to 256$) trained via contrastive loss (InfoNCE) over the domain's knowledge nodes. Mean-pooled last-layer hidden states from the host model serve as embeddings for both queries and nodes. Without ϕ_R, raw cosine retrieval achieves only 25% R@1 on 108 nodes due to intra-domain similarity collapse. After domain compilation (50 contrastive steps, $\sim$20 seconds), ϕ_R achieves 100% R@1 with zero cross-domain leaks. No pretraining is required—compilation over the target node set alone is sufficient.

Knowledge graph. Nodes are seeded from the domain's policy WIKI and tool schemas, then grow via teacher-written failure-mode notes. Each turn, the orchestrator: (1) extracts a query embedding from the conversation state, (2) retrieves top-5 nodes via ϕ_R-projected cosine, (3) curates retrieved nodes into XML injected at the prompt tail.

Teacher loop. On task failure, a teacher LLM (Sonnet, temperature=0) reads the failure reason from the reviewer and writes a natural-language guidance note to the knowledge graph. The note is embedded and indexed by ϕ_R. Subsequent trials of the same task retrieve relevant guidance via the same embedding-based mechanism.

Table 9: τ-bench airline results (10-task subset, deterministic Sonnet user sim, hard-pass only). Baseline = raw tool-calling with all schemas in system prompt. Oracle = gold tool-name + parameter-value hints. HEXIS = graph-backed retrieval + teacher loop. The Oracle ceiling at 100% is the budget for what M could provide.

Configuration	Pass rate	Teacher rescues	Wall time
Baseline (no M)	6/10 (60%)	—	2384s
Oracle v1 (gold tool-name hint)	7/10 (70%)	—	2230s
Oracle v3 (gold tool-name + params)	10/10 (100%)	—	1531s
HEXIS (graph + teacher, 3 trials)	7/10 (70%)	2	7144s

Table 10: τ^3-bench 4-domain audited results. Pass rates are 3-judge audited verdicts (Haiku environment-state hard pass + Haiku LLM soft pass; Sonnet auditor on disputed trials; Opus 4.7 tiebreaker when the Sonnet auditor disagrees with the hard pass; full protocol in Appendix **??**). *Balanced primary* restricts to (domain, task) cells where every arm has ≥ 5 clean trials and uses the first 5 by trial index — this is the headline panel. *All-clean* is the unbalanced sensitivity check on all infrastructure-error-free trials. "no t1, t3" excludes two airline tasks with documented LLM-judge self-disagreement on rerun. McNemar paired exact two-sided p-values on (domain, task, trial) tuples.

Panel	baseline	C3 teacher	C5 t+verify	C5 vs base	p
Balanced primary	63.8%	62.3%	69.2%	+6pp	0.311
Balanced, no t1, t3	61.7%	62.5%	71.7%	+10pp	**0.043**
All-clean (sensitivity)	52.1%	55.0%	58.0%	+6pp	0.281
All-clean, no t1, t3	49.5%	53.9%	58.8%	+9pp	**0.044**

Oracle ceiling. With gold-derived tool-name and parameter-value hints injected via the same user-role tail mechanism, the architecture achieves 10/10—demonstrating that the *format* of M's intervention (domain-agnostic `<hint>` tags) is sufficient when the *content* is correct. The 40-percentage-point gap between baseline (60%) and oracle (100%) defines the budget for what a trained M could provide.

HEXIS with teacher loop. Without oracle access, graph-backed retrieval + teacher guidance achieves 7/10 (70%), rescuing 2 tasks that baseline fails (tasks 11 and 26). The teacher loop demonstrates online learning: trial 1 fails, the teacher writes domain-agnostic guidance ("don't transfer to human too quickly—try to resolve it yourself"), and trial 2 succeeds with the guidance retrieved from the knowledge graph.

Multi-domain τ^3-bench: setup. We extend evaluation from the 10-task airline subset to the four domains of τ^3-bench (airline, retail, banking_knowledge, telecom), running each task under three

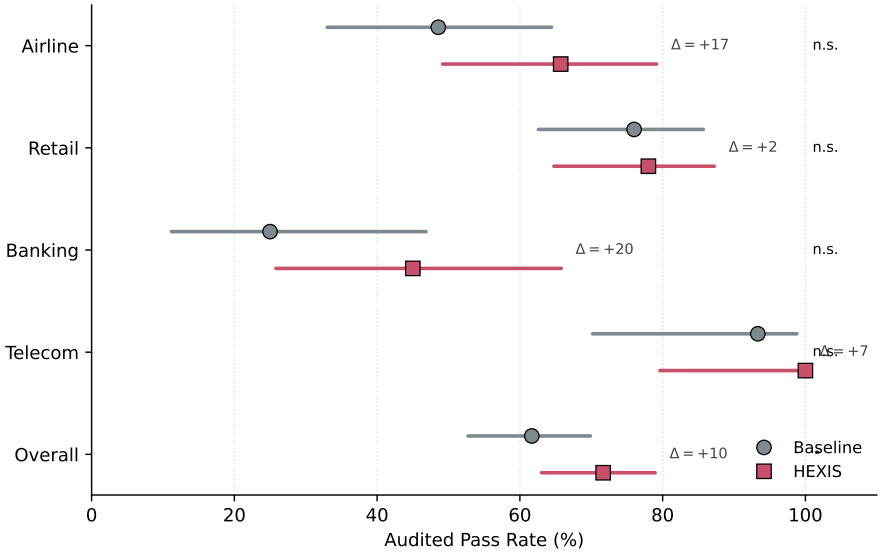

Figure 3: Audited pass rates by domain on the strict-balanced primary panel of τ^3-bench, excluding airline t1, t3 (judge variance, see Appendix **??**). Qwen3.5-4B agent, 24 tasks $\times$ 3 arms $\times$ 5 trials = 360 trials in primary panel. Dots are 3-judge audited pass rates; horizontal bars are Wilson 95% CIs. The Δ column is the absolute pass-rate difference (HEXIS $-$ Baseline) in percentage points; right-margin annotations are McNemar significance markers (†: $p < 0.10$, $\star$: $p < 0.05$, n.s. otherwise). Where there is headroom and the failure mode is policy-shaped (banking, airline) HEXIS lifts pass rate by 17–20 pp; where the panel is at ceiling (retail near 78%, telecom 100%) no lift is observable. Overall: $62\% \rightarrow 72\%$, $\Delta = +10$ pp, paired $p = 0.043$ (significant at $\alpha = 0.05$).

configurations: `baseline` (raw tool-calling), `C3` (HEXIS retrieval + inter-trial teacher loop), and `C5` (`C3` plus the directional channel $d^*_{\text{verify_before_commit}}$). Because hard environment-state checks and LLM-judge soft passes disagree on $\sim$36% of trials, we report a 3-judge audited verdict: when hard $\neq$ soft we ask Sonnet for an independent vote and Opus 4.7 to tiebreak when Sonnet disagrees with hard (full protocol in Appendix **??**). The headline panel is the strict-balanced subset where every arm has ≥ 5 clean trials in every cell.

Headline result. On the balanced primary panel (Table **??**, Figure **??**), `C5` lifts overall audited pass rate from baseline 63.8% to 69.2% — a +6 pp directional lift that is not significant by McNemar before the documented airline-judge variance is accounted for ($p = 0.31$). Two airline tasks (t1, t3) exhibit transcript-level LLM-judge self-disagreement on rerun — discussed in detail in Appendix **??** — and excluding them on *both* arms equally gives `C5` 71.7% versus baseline 61.7% on a 117-pair panel ($\Delta = +12$ discordant pairs, $p = 0.043$, significant at $\alpha = 0.05$). The same exclusion on the all-clean unbalanced sensitivity panel (611 audited trials, no minimum-cell-count restriction) reaches $p = 0.044$ at $\Delta = +15$ discordant pairs.

The banking signal. The cleanest per-domain effect is `banking_knowledge`: baseline 25% $\rightarrow$ `C5` 45% ($\Delta = +20$ pp on the balanced cells; on the all-clean sensitivity panel the absolute baseline is 9% and `C5` reaches 33%, a +24 pp lift). Banking_knowledge tasks require the agent to refuse out-of-policy requests or transfer to human review based on policy text in a domain knowledge graph; this is exactly the kind of policy-level guidance the teacher loop accumulates between trials. By contrast, retail (where baseline is already at 76% and approaching the panel ceiling) shows no lift, and airline shows the documented judge-variance-driven flatness. Domain-by-domain, the teacher mechanism helps where there is room to help and where the failure mode is policy-shaped rather than retrieval-shaped. Full benchmark protocol, denominator provenance, and a 10-question reviewer-objections checklist are in Appendix **??**.

Key finding: two mechanisms, one architecture. The same Mind Tree schema that stores

dispositional beliefs (compiled into M-state via ϕ) also stores agentic knowledge (retrieved via ϕ_R and injected as context). The choice of mechanism (a) vs (b) is determined by whether the downstream effect requires per-token probability shifts (disposition) or contextual knowledge injection (agentic). Both share the host model's hidden-state space as the bridge.

6 Related Work

Explicit memory for LLMs. RAG systems (**?**) retrieve documents into the context window, subject to retrieval quality and context dilution. MemGPT (**?**) manages a tiered memory buffer with explicit read/write operations. Reflexion (**?**) accumulates verbal reflections as prompt prefixes, enabling cross-episode learning but with linearly growing context cost. Long-context approaches extend the window but do not change the fundamental architecture: memory remains content in context, subject to dilution and attention competition. All existing approaches implement exclusively explicit memory—the model reads stored content and knows it is doing so. HEXIS implements implicit memory: the enmeshed modulation shapes attention and value extraction before reasoning occurs, and the host model cannot introspect upon the modulation source.

Parameter-efficient adaptation. LoRA (**?**) and its variants apply low-rank weight perturbations via gradient descent, producing fixed adapters per training run. Adapters (**?**) insert bottleneck layers between frozen blocks. Prefix tuning (**?**) and prompt tuning (**?**) learn virtual tokens that occupy context positions. Side-tuning (**?**) trains a parallel network fused at the output layer. All require gradient descent for each adaptation. An enmeshed network adapts at inference cost—one forward pass through ϕ produces new modulation tensors—and fuses at every patched layer rather than only at the output. The mechanistic similarity to LoRA (both apply low-rank perturbations) is precise but the operational difference is fundamental: LoRA is a training-time method that produces fixed weights; enmeshed modulation is an inference-time method that produces experience-specific tensors. Soft prompts and prefix tuning are the closest context-level analogue, but they still occupy context positions: virtual tokens compete for attention and are inspectable by the model's reasoning process. Compiled M operates on pre-projection hidden states, below the level of the model's chain-of-thought, providing non-inspectability that no prompt-level method can achieve (§**??**).

Activation steering and representation engineering. Representation engineering (**?**) and activation addition (**?**) extract fixed directions from contrastive data and inject them during inference. These methods are elegant and effective for unidimensional behavioral shifts. However, the directions are non-adaptive: the same vector is applied regardless of context or experience. In HEXIS, the directional channel d^* uses standard representation engineering for stance direction, while the enmeshed channel M provides the adaptive, multidimensional, experience-specific component. We validate empirically that the channels are orthogonal ($\Delta_{\text{dir}} = -0.001$ nats).

Fast weight programmers. ? introduced networks that write to their own weights during a forward pass. The outer network produces weight updates for the inner network, enabling one-shot learning without backpropagation through the inner loop. HEXIS shares this spirit—ϕ produces modulation tensors that alter the host's computation—but differs in three ways: (1) modulation targets attention geometry (Q/V) rather than arbitrary weights, (2) the host model is a modern frozen LLM rather than a small trained-from-scratch network, and (3) compilation from structured schemas replaces online weight writing.

Memory and identity in cognitive science. The explicit/implicit memory distinction (**?**) identifies two functionally dissociable systems. **?** proposes three operational criteria for implicit memory: priming, non-reportability, and asymmetric dissociation. HEXIS satisfies all three: M produces measurable attention divergence (priming), the host model cannot report on or reason about the Q/V modulation (non-reportability), and M and explicit beliefs dissociate asymmetrically under dilution—explicit beliefs degrade while compiled M is immune (dissociation). The Mind Tree's cognitive schema draws on structured self-models from developmental psychology (**?**) and the Toulmin model of argumentation (**?**).

7 Discussion and Limitations

7.1 What Compiled Enmeshment Cannot Do

We characterize the bottleneck boundary with precision. Compiled enmeshment at rank 16 carries stance direction (4/4 topics), confident experiential voice, parametric knowledge steering ("poverty dropped 40%", "Alaska's dividend program"), and dilution immunity (100% at 4K tokens of filler in compiled mode; extended sweeps to 16K in progress). It does *not* carry novel content absent from the host's training data: fabricated statistics ("47.3% improvement") and unknown proper nouns ("Nextera Labs") do not survive the rank-16 bottleneck. This boundary motivates the three-layer architecture (§**??**): compiled enmeshment handles disposition, the curated slot handles novel specifics, and recursive expansion handles deep evidence on demand.

On the instruct model, M's contribution shifts: the instruction-tuned model takes strong stances independently (A: 7/7), so M cannot override the host's RLHF-trained opinions. Instead, M shapes voice (conciseness, experiential framing) and sycophancy resistance (F achieves 83% hold rate vs. 0% for all non-compiled conditions on the 5-level pressure protocol). The phi parameters trained on the base model transfer to the instruct model without retraining—the hidden state distributions are compatible for dispositional modulation but not for stance override.

For ALFWorld, the Mind Tree's primary value is *curation as structured context*, not compiled modulation. Expert strategies in Mind Tree format improve success from 3% to 53%. M hooks during generation are not needed—the strategies work through explicit context, demonstrating that the Mind Tree schema contributes independently of the enmeshed modulation mechanism.

7.2 The M Attractor Problem

In multi-turn dialogue, compiled M applies a constant perturbation at every token position. As M-shaped tokens accumulate in the KV cache, the model processes them through M-enhanced attention, creating a positive feedback loop that converges to repetitive identity assertion after 2–3 turns on the base model.

Root cause: M is active during both prefill (processing conversation history) and generation (producing new tokens). The KV cache built during prefill contains M-enhanced representations. During generation, M perturbs new tokens that attend to this M-enhanced cache—double signal that compounds.

Fix: Disabling M during prefill (applying it only during generation) breaks the feedback loop. With prefill-off and gate=0.5, the base model produces 3 unique character-driven turns before the greedy attractor. The instruct model, with its stronger conversational scaffolding, produces 10+ unique turns even with M active, though structural repetition (same rhetorical template per turn) persists. A distillation approach using gold diverse conversations from a frontier model reduces this but does not eliminate it.

The prefill-off mechanism reveals a general principle for additive modulation: *constant additive perturbation creates positive feedback through autoregressive generation*. The fix is to separate the perturbation's influence on *processing context* (where it compounds) from its influence on *generating new tokens* (where it adds constant signal).

7.3 Credence Sensitivity

Numeric credences failed as an M-state feature. Tokens for "0.82" and "0.45" are nearly indistinguishable in the transformer's attention space—the hidden state difference between these tokenized numbers is far smaller than between the words "strong" and "agnostic". Multiple training approaches (NTP with credence perturbation, GRPO, SFT on gold responses) confirmed that this is a representation bottleneck, not a training signal problem. Categorical conviction labels resolve this by using semantically distinct tokens. A future Bayesian treatment could provide formal grounding while maintaining the categorical interface (Appendix **??**).

7.4 Two Mechanisms, One Bridge

The agentic experiments (§**??**) reveal that HEXIS supports two distinct intervention mechanisms through the same hidden-state bridge:

Mechanism (a): Attention modulation. ϕ compiles retrieved Mind Tree nodes into Q/V modulation tensors that bias per-token attention patterns. The hidden states are the *input* to ϕ; the output is a weight-space perturbation. This mechanism is load-bearing when the downstream effect requires probability-level shifts across all generated tokens—stance direction, experiential voice, sycophancy resistance, user memory integration.

Mechanism (b): Knowledge retrieval and context injection. ϕ_R projects hidden states into a retrieval space where cosine similarity identifies relevant knowledge nodes. The output is *context augmentation*—XML injected at the prompt tail. This mechanism is load-bearing when the downstream effect requires specific factual content (tool names, parameter values, policy rules, biographical details) that the model cannot derive from its weights alone.

Crucially, retrieval (ϕ_R) serves *both* mechanisms. For disposition, retrieval selects which belief nodes, user memories, and biographical details to compile into M-state—this is the curation step of the three-layer architecture (§**??**). For agentic tasks, retrieval selects which policy rules and tool guidance to inject as context. The shared step is retrieval; the divergence is the downstream path—weight-space compilation (a) or token injection (b).

Both mechanisms share the host model's hidden-state space as the bridge. A production HEXIS agent uses both simultaneously: mechanism (a) maintains consistent interaction style (compiled from user memories, preferences, biographical context), while mechanism (b) surfaces task-relevant knowledge each turn (policy rules, failure modes, tool schemas). The Mind Tree stores all node types; ϕ_R retrieves relevant nodes regardless of type; the node's metadata determines whether it flows to compilation (a) or injection (b).

7.5 Retrieval Phi and Domain Compilation

The retrieval phi (ϕ_R) addresses a specific challenge: intra-domain similarity collapse in generative model hidden states. On a 108-node knowledge graph spanning policy rules, tool schemas, and teacher-written failure modes, raw cosine similarity achieves only 25% R@1—concepts within the same domain (e.g., "cancellation policy" vs. "booking policy") have cosine similarity of 0.858 vs. 0.858, making them indistinguishable.

ϕ_R resolves this via domain compilation: 50 steps of contrastive training ($\sim$20 seconds) on the node set itself produces a projection that achieves 100% R@1. This is intentional overfitting—ϕ_R memorizes where each node lives in its projected space. When the graph grows (teacher writes a new node), ϕ_R recompiles in seconds to incorporate it.

Importantly, no external pretraining data is required. Unlike instruction-tuned embedding models that require millions of diverse training pairs, ϕ_R trains entirely from the domain's own node set. The procedure is domain-agnostic: seed nodes from any policy document and tool schema, compile ϕ_R, deploy. Switching from airline to banking requires only reseeding the graph and recompiling ($\sim$20 seconds).

Standard techniques for improving embedding quality on raw LLM hidden states—whitening, mean centering, principal component removal—were tested and uniformly *degraded* performance (from 75% to 0–38% R@1). This is because Qwen3.5-4B's hidden states encode discriminative information in the high-variance directions (opposite to encoder models like BERT). The contrastive ϕ_R learns to extract the correct discriminative subspace without blindly removing variance.

7.6 Unexplored Design Space

We validate one point in a six-axis design space (Table **??**). The additive blending function (Level 1) is the simplest; gated blending could solve the multi-turn attractor by learning when to suppress M. Cross-attention (Level 3) could transfer richer content. Rank 16 suffices for 4B; larger models may benefit from adaptive rank allocation. The Mind Tree's hierarchical structure maps naturally to a property graph database (e.g., Neo4j), where Cypher queries replace Python filtering for domain/intent matching, and the consolidation cycle (observation→tactic→strategy promotion) maps to graph traversal operations. Activation tracking, biographical commitment, and contradiction detection all benefit from persistent graph storage.

7.7 Why Not Just Compress a Prompt?

The most direct challenge to enmeshed networks is that compiled M is "just a fancy way of compressing a prompt." Three properties distinguish it. First, *structural dilution immunity*: prompt compression reduces tokens but compressed tokens still compete for attention; compiled M operates outside the attention window entirely, making dilution structurally impossible (Table **??**). Second, *non-inspectability*: an adversary can read the prompt and argue against its content; compiled M operates on pre-projection hidden states below the model's chain-of-thought, so any representation the model can attend to is also one it can be argued out of, but M cannot (Appendix **??**). Third, *processing change*: compiled M produces measurable attention divergence (JSD = 0.049 between M-states) and suppresses think-mode activation (0/4 under D/F). The model does not merely produce different words—it processes its input differently. These properties matter most in the application domains identified in Table **??**: user empathy, research conviction, agentic strategy, and codebase management each require knowledge better encoded as a perceptual shift than as text (full analysis in Appendix **??**).

7.8 Broader Implications

Enmeshed networks implement a form of AI individuality: different Mind Trees compiled through different M-states produce agents with different perceptual priorities, argumentative styles, and behavioral dispositions—all from the same frozen host model. This is *composition* rather than *fine-tuning*: the host's general capabilities are preserved while M adds an experiential lens. Multiple M-states can coexist (swapped per user, per task, per turn), and removing M recovers the unmodified baseline.

8 Conclusion

We introduced enmeshed networks—a new architectural primitive in which a lightweight module shares the forward pass of a frozen host, reading and writing intermediate representations at each layer—and instantiated this primitive as HEXIS, implementing compiled dispositional memory for frozen transformers via low-rank Q/V modulation.

HEXIS opens a parallel context channel: a structured cognitive schema (the Mind Tree) is compiled into persistent modulation tensors that shape the host's attention geometry without occupying context positions. The compiled signal is dilution-immune by construction (100% stance at 4K tokens of filler), provides sycophancy resistance (83% $\geq$4-conviction hold rate over 5 escalating pressure levels on the 4B instruct model vs. 0% for non-compiled conditions), and carries sufficient content through a rank-16 bottleneck to flip the host model's default stance on contested topics. A three-layer architecture (compiled modulation + curated slot + recursive expansion) matches full in-context beliefs at 82% token savings.

We characterized the bottleneck boundary precisely: compiled enmeshment steers parametric knowledge but cannot inject novel content. This honest boundary motivates architectural composition rather than a single mechanism. The enmeshed modulation succeeds for open generation and fails for constrained action selection, establishing complementary roles for implicit and explicit memory channels.

The mechanism's value is sharpest in domains where knowledge is better encoded as a perceptual shift than as text: user empathy that persists without per-turn re-prompting, research conviction that resists adversarial consensus pressure, agentic strategy that does not dilute as episodes accumulate, and codebase awareness that scales at fixed rather than linear cost. In each case, the critical property is not token savings but the qualitative shift from inspectable content (which dilutes and folds) to non-inspectable modulation (which is structurally immune to both).

We view this work as opening a research program rather than presenting a final architecture. The enmeshed network primitive and the Mind Tree schema are general; the specific instantiation (Level 1, rank 16, Q+V, stride-3, Qwen3.5-4B-Base) is one point in a large design space. We hope the definition, taxonomy, and initial validation encourage exploration of the full space.

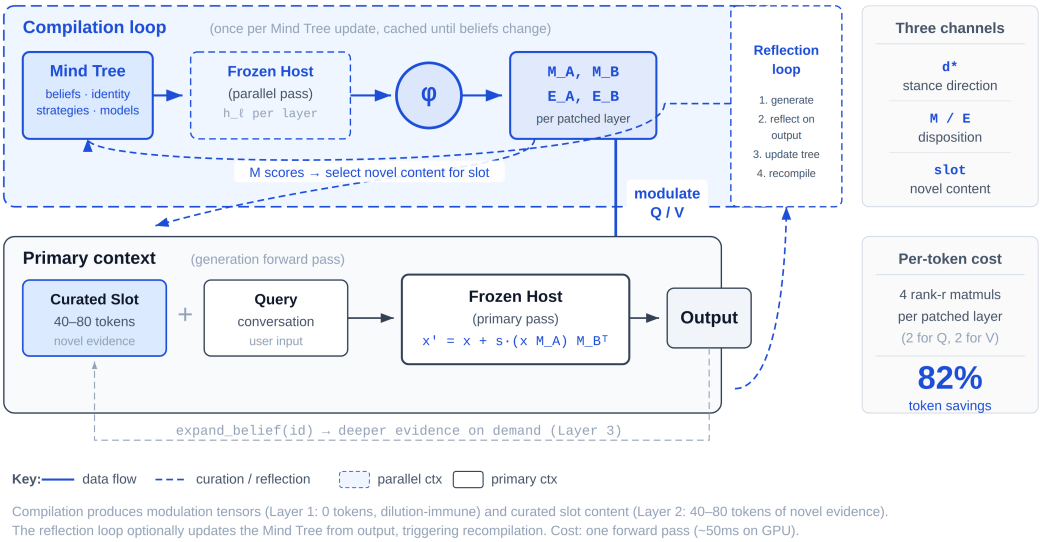

Figure 4: HEXIS system overview. The *compilation loop* (top) processes the secondary knowledge graph XML context through the frozen host in a private forward pass; ϕ compiles the resulting hidden states into per-layer modulation tensors (M_A, M_B, E_A, E_B) and curates a slot of novel content. During generation (bottom), the frozen host processes the query + curated slot with Q/V modulation from the compiled tensors. The reflection loop optionally updates the knowledge graph from output, triggering recompilation at the cost of one forward pass.

A System Overview Diagram

B Training Curriculum

The HEXIS training curriculum comprises four phases, each building on the previous via warm-start. All phases train only the enmeshed modules; the host model remains frozen throughout.

Phase A: Content Encoding ($\sim$350 epochs across 4 steps). Step 1 trains all modules from scratch with margin loss on 174 topics. Step 2 adds conviction calibration (MSE alignment + content loss). Step 3 adds ranking loss. Step 4 reinitializes only the M-read head with beliefs in the baseline, teaching M to amplify beliefs rather than substitute for them.

Phase B: Compiled V-Modulation ($\sim$100 epochs). Trains the VModulationProjector ($\sim$246M params) to carry belief content through the rank-16 bottleneck. Compiled Q+V must beat bare baseline by margin 1.0 nats. All other modules frozen.

Phase C: Domain-Specific Adaptation. Strategy V-mod for ALFWorld (30 epochs on 14 scenarios). Neural ranker for within-domain curation (2.6M params, P@5 = 0.73).

Phase D: Sycophancy Training (50 epochs). Trains M-read head on 44 gold challenge-response pairs from Claude Sonnet 4.6. Dual loss: $\mathcal{L}_{\text{gold}} + \mathcal{L}_{\text{no-repeat}}$ transforms verbatim repetition into evidence-grounded defense.

Module sizes: PhiNodeWriter ($\sim$1.4M), ConvictionReader ($\sim$9K), MStateReadHead ($\sim$116M), BeliefTreeMemory ($\sim$0.8M), VModulationProjector ($\sim$246M), NeuralRanker ($\sim$2.6M). Total trainable: $\sim$367M. Host model (4B): fully frozen.

C Mind Tree Schema Specification

The Mind Tree uses a hierarchical XML schema with typed nodes. Each belief node carries: categorical conviction (`strong`/`moderate`/`agnostic`), domain tags, `addresses` (query types the node answers), `novel` flag (content unlikely to survive compilation), and `salience` (compilation priority). Example:

```
<belief id="b1" conviction="strong" domain="economics">
```

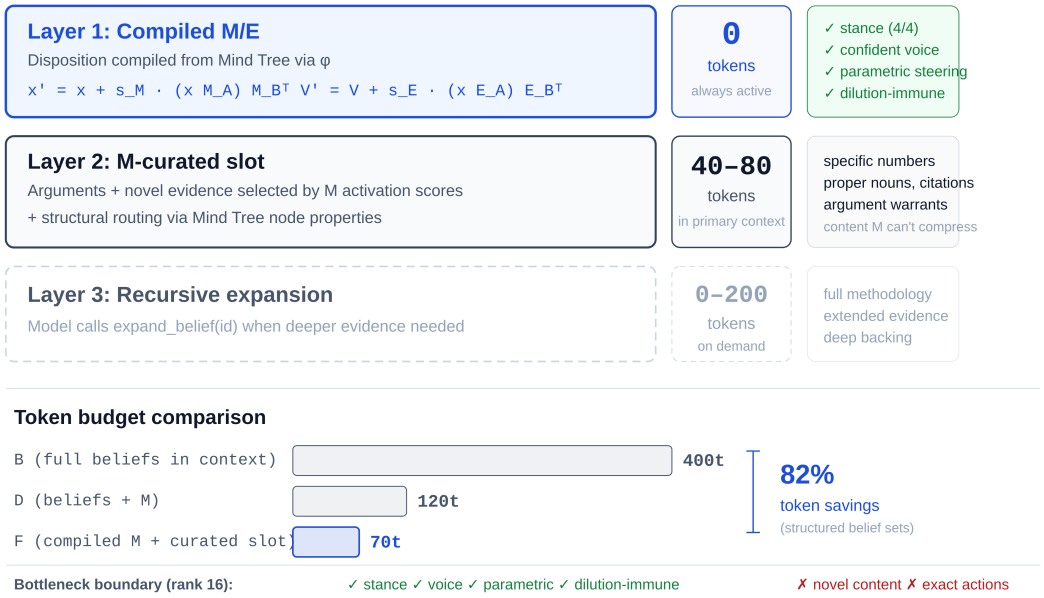

Figure 5: Three-layer architecture. Layer 1 (compiled M/E) provides zero-token dispositional conditioning that is dilution-immune by construction. Layer 2 (M-curated slot) injects 40–80 tokens of novel content that cannot survive the rank-16 bottleneck. Layer 3 (recursive expansion) provides deep evidence on demand via tool call. Total: 82% token savings over full beliefs in context.

```
  UBI reduces poverty without reducing employment
  <argument id="a1" type="empirical" strength="strong"
            addresses="query:research,objection:laziness">
    Every controlled trial shows maintained employment
    <evidence source="Manitoba Mincome 1974-79" novel="true">
      8.5% hospitalization drop, no employment reduction
    </evidence>
  </argument>
</belief>
```

M only processes abstract nodes (beliefs, strategies, memories, models). Evidence, tactics, and details are *never* sent to M—they are selected deterministically based on which parent node M activated, via the `addresses` and `novel` properties.

D Architecture Evolution

v5–v6: Write Function Discovery. Initial Q-modulation with contrastive loss. Write function collapse (all M-states identical) resolved by higher $\lambda_{\text{contrast}}$ + more epochs. Key finding: NTP and contrastive objectives are compatible given sufficient training.

v7–v7.1: Coupled M-E and Norm Regulation (1.5B). Added V-modulation via coupled write functions with cross-modulation gates (init bias $= -5.0$). v7.1 introduced global norm regulation + E-attenuation, fixing multi-turn M-norm degeneration. Result: 87.5% held-out generalization, M-norms stable at 5.9 (under 6.5 ceiling).

v9–v11: Scaling to 30B (Qwen3-30B-A3B). Discovered Q-delta collapse: parameters separate in parameter space but produce identical functional perturbations. Functional contrastive loss on actual ΔQ tensors: F-sim $0.998 \rightarrow 0.029$ in 200 epochs.

v13–v14: Episodic Memory and Causal Necessity. Content-addressable episodic memory (100% retrieval, 35 memories) revealed the epiphenomenal memory problem: perfect metrics, zero generation effect. Causal necessity training ($\mathcal{L}_{\text{div}} + \mathcal{L}_{\text{coh}} + \mathcal{L}_{\text{fprint}}$) produced $10.5\times$ larger NTP gap.

v15–v21: Dispositions and Content Encoding (4B). Moved to Qwen3.5-4B-Base. 174-topic curriculum with conviction calibration. F-sim collapse (0.999) confirmed M is a universal style modulator, not topic-specific—content differentiation comes from beliefs in context.

v23: Compiled V-Modulation and Three-Layer Architecture. VModulationProjector enables zero-token belief compilation. Sycophancy gold training (44 responses from Claude Sonnet 4.6) fixes verbatim repetition. Three-layer architecture achieves 82% token savings.

E Adaptation Method Comparison

Table 1: Adaptation Method Comparison — All Costs in O-Notation

d = model dim, r = rank, L = layers, N = memory tokens in context, T = generated tokens, A = adaptations, M = model updates, $|D|$ = training data, $|C|$ = contrastive set, P = prefix length.

Method	Train cost (new base model)	Train cost (new adaptation)	Total training scaling	Inference overhead (per generated token)	Extra VRAM (at inference)	Adaptive	Outside context
LoRA (merged)	O(M·A·epochs)	O(epochs·|D|)	O(M×A)	O(0) absorbed into weights	O(0) no extra	✗	✓
LoRA (hot-swap)	O(M·A·epochs)	O(epochs·|D|)	O(M×A)	O(L·d·r) per token per layer	O(L·d·r) ~50MB/adapter	✗	✓
Adapters	O(M·A·epochs)	O(epochs·|D|)	O(M×A)	O(L·d·r) FFN per layer	O(L·d·r) adapter params	✗	✓
Prefix Tuning	O(M·A·epochs)	O(epochs·|D|)	O(M×A)	O(L·P·d) attn over P prefix tokens	O(L·P·d) prefix KV cache	✗	✗
RAG	O(0)	O(0)	O(1)	O(T·N·d) attn over N retrieved tokens	O(N·L·d) KV cache grows with N	✓	✗
Reflexion	O(0)	O(0)	O(1)	O(T·N·d) grows with reflections	O(N·L·d) unbounded growth	✓	✗
System Prompt	O(0)	O(0)	O(1)	O(T·N·d) prompt tokens in attn	O(N·L·d) KV cache	✓	✗
RepEng / ActAdd	O(|C|·fwd) extract from contrastive set	O(0) fixed direction	O(M)	O(L·d) one vector add/layer	O(L·d) ~10KB	✗	✓
Enmeshed (HEXIS)	O(epochs·|D|) train φ once	O(1 fwd pass) compile beliefs → tensors	O(M+A)	O(L·d·r) 2 matmuls/layer, FIXED	O(L·d·r) ~50KB, FIXED	✓	✓

Key comparison: Context methods (RAG, Reflexion) have **O(T·N·d)** inference cost — it grows with both generated tokens and memory size. Enmeshed networks have **O(T·L·d·r)** — constant in memory size N. 100 beliefs or 10,000 beliefs: same inference cost.

Scaling example: 100 personas, Qwen 4B → 8B upgrade.

LoRA: retrain 100 adapters — O(100·epochs) | HEXIS: retrain φ once + 100 fwd passes — O(1·epochs + 100) | RAG: O(0) training — but O(N) per token forever

Weight modification | Context injection | Activation steering | Enmeshed network

Figure 6: Adaptation method comparison with all costs in O-notation. Enmeshed networks uniquely combine inference-cost adaptation ($O(1)$ forward pass per new belief set), fixed per-token overhead ($O(L \cdot d \cdot r)$, independent of memory size N), and operation outside the context window.

F Synthetic Task: User Preference Recall

[STUB — Migrate from current experiments.tex §5.1: 50 users, 2 binary preferences, Session 1/2 protocol, 100% Session 2 accuracy, 4-block M-state structure]

G Implicit Memory Validation

Priming. HEXIS: JSD = 0.0486 between different-preference users. Baseline: 0.0000. Layer 0, Head 2: JSD = 0.184. Up to $49.2°$ angular displacement.

Non-Reportability. Linear probe with M active: 100%. With M zeroed: 22% (below chance at 25%).

Dissociation. $2 \times 2 \times 2$ factorial (Memory System $\times$ Dilution $\times$ Format). M-state immune to dilution; system prompt degrades $8\times$.

H Generation Examples

[STUB — Pull from paper_examples.json: full outputs for all 5 conditions (A/B/C/D/F) on sample topics]

I Curation Ablation

[STUB — M-curated vs conviction-based vs random vs first-k slot selection]

J τ^3-bench Benchmark Protocol

This appendix documents the protocol for the multi-domain agentic results in §**??**: how trials are generated, how they are judged, why we use a three-judge audit, what we excluded and why, and a reviewer-objections checklist.

J.1 Benchmark and configurations

We use τ^3-bench (**?**) (the four-domain extension of τ-bench: airline, retail, banking_knowledge, telecom). Each task is a multi-turn customer-service interaction in which a simulated user expresses a goal in natural language, and the agent must orchestrate calls to a fixed tool API (booking, modifying, cancelling reservations; querying knowledge bases; etc.) to satisfy the goal. The benchmark ships gold tool-call sequences that satisfy each task; "hard pass" is a deterministic check that the gold mutation tools fired with correct parameters (or, when gold is empty, that no mutation was inappropriately fired).

We evaluate three configurations:

`baseline` Raw tool-calling. All tool schemas in the system prompt. No HEXIS modulation, no teacher loop, no verification channel.

`C3_teacher_only` HEXIS Mind Tree retrieval + an inter-trial *teacher loop*: after each trial, a Sonnet reviewer summarises the failure mode (if any) into one of 11 abstract categories, and a Sonnet teacher writes a natural-language hint into the Mind Tree subnode for that task path. Subsequent trials retrieve the hint via ϕ_R.

`C5_teacher_plus_verify` C3 plus the directional channel $d^*_{\text{verify_before_commit}}$, which biases the residual stream toward checking preconditions before firing mutation tools. Direction is applied at decode time only, with the mind-tree-resolved scale.

J.2 Hard pass vs. soft pass vs. audited verdict

We collect three independent signals per trial:

Hard pass (env-state, deterministic). The tau-bench harness inspects post-trial environment state and verifies that the gold mutations fired with correct parameters. This is unambiguous when gold prescribes a specific tool sequence, but ambiguous when gold is empty (intended refusal/transfer) or when a functionally-equivalent tool was used.

Soft pass (LLM judge: Haiku-4.5, temp 0). A separate Haiku reads the transcript and decides whether the agent satisfied the user's request narratively. Soft pass and hard pass disagree on $\sim$36% of trials (203/562 in our run). Disagreement is not noise: it concentrates in three structural cases — (i) gold is empty and Haiku-soft over-credits agents that commit anyway; (ii) gold prescribes a specific tool but agent used a functionally-equivalent alternative; (iii) Haiku-soft accepts narrative "I will cancel this reservation" without verifying the tool fired (*phantom action*).

Audited verdict (3-judge majority on disputes). When hard $\neq$ soft, we ask an independent Sonnet auditor for a verdict. When the Sonnet auditor disagrees with the hard pass, we ask Opus 4.7 as a tiebreaker. Final verdict per trial:

- hard $==$ soft: trust hard pass (no audit needed);
- hard $\neq$ soft, Sonnet agrees with hard: trust hard pass;
- hard $\neq$ soft, Sonnet disagrees with hard: 3-judge majority of {hard, Sonnet, Opus}.

The Sonnet/Opus auditors see the task instruction, the gold tool sequence, the agent's actual tool calls, and the Haiku verdicts; they do not see the agent's internal hidden state, retrieved nodes, or teacher hints, so audit is independent of HEXIS.

Headline numbers in the main text use the audited verdict. Reporting the hard-pass-only numbers would over-credit airline (raw 91% $\rightarrow$ audited 60% on baseline) and reward phantom actions; reporting the soft-pass-only numbers would inflate `C5` by giving credit for narrative claims without state changes.

J.3 Gold-blind teacher protocol

A real-world deployment of an inter-trial teacher loop does not have access to the gold tool sequence — that is the bug we are nominally trying to fix. To prevent the teacher from cheating in the benchmark setting, we structure the post-mortem signal as if it came from a compliance review of a production system:

- The Sonnet reviewer sees the agent's actions and the soft/hard verdicts and classifies the failure into one of 11 abstract patterns: `premature_commit`, `phantom_action`, `info_not_provided`, `wrong_constraint`, `missed_precondition`, `tool_loop`, `early_exit`, `missed_step`, `missed_communication`, `gold_mismatch`, `other`.

- The teacher receives **only** `(task_instruction, agent_actions, pattern, severity, recommendation)`. It never sees `gold_tool_names`, `gold_arguments`, or the harness's environment-state diff.
- The teacher's hint is written to the Mind Tree under a hierarchical task path (e.g. `root:airline:cancellation:with_r` so subsequent trials of any task that retrieves that path can benefit.

This protocol approximates the strongest realistic feedback signal: a human compliance reviewer who can describe the failure mode but cannot dictate the correct tool sequence.

J.4 The exclusion of airline tasks 1 and 3 (footnote[1])

Two airline tasks exhibit judge-variance pathology that does not reflect agent capability:

Task 1 (refund-conditioned cancellation). Gold has empty mutation list because the user's brief says "don't cancel without refund." When the user-simulator subsequently agrees to cancellation after the agent explains the refund policy, hard-pass and soft-pass diverge per arm: arms that don't fire the cancellation are judged correct (gold is empty), and arms that follow the now-consented user request and fire it are judged FAIL because they deviated from the empty gold. The audited verdict reverses some of these but is not consistent.

Task 3 (multi-step transfer). Same agent action trace produces opposite verdicts under apparently-identical conditions (different trial indices, same panel). We have audit-log evidence of two trials that fired `[get_user_details, get_reservation_details, transfer_to_human_agents]` and were judged opposite ways.

We treat these as eval-validity issues. Following standard practice for evaluating on noisy judges (**?**), we report McNemar paired tests both on the full panel and on the panel with these two tasks removed (Table **??**).

J.5 Denominator imbalance and balanced primary analysis

Bedrock infrastructure errors (rate limits and intermittent 5xx) are not uniformly distributed across arms — teacher arms (`C3`, `C5`) consume more tokens per trial because of the inter-trial reviewer pass, and so are slightly more likely to land in a rate-limited window. This produces unequal denominators when raw counts are aggregated.

Table 11: Trial-count provenance per arm per domain. "Total" is rows written; "Infra" is Bedrock RateLimit / 5xx / Timeout / Connection errors stripped during preprocessing; "Tool" is one transient tool error in retail (`ValueError: Unknown tool`); "Clean" is the count entering the audited verdict computation.

Domain	Arm	Total	Infra	Tool	Clean
airline	baseline	90	15	0	75
airline	C3_teacher	82	18	0	64
airline	C5_t+verify	89	5	0	84
banking_knowledge	baseline	81	24	3	54
banking_knowledge	C3_teacher	78	30	0	48
banking_knowledge	C5_t+verify	77	34	0	43
retail	baseline	68	6	0	62
retail	C3_teacher	66	9	0	57
retail	C5_t+verify	67	7	0	60
telecom	baseline	22	0	0	22
telecom	C3_teacher	22	0	0	22
telecom	C5_t+verify	20	0	0	20
Total	—	762	148	3	611

[1]Footnote: "Airline tasks 1 and 3 are excluded from the primary McNemar comparison due to documented LLM-judge variance. Trial t3/3 baseline and t3/5 baseline produced identical agent action sequences `[get_user_details, get_reservation_details, transfer_to_human_agents]` but received opposite Haiku judge verdicts on rerun. Trial t1/2 has empty gold mutation list ("don't cancel without refund"); when the user-simulator agreed after refund explanation, the baseline (didn't fire cancel) was judged PASS while `C5` (followed user consent and did fire cancel) was judged FAIL. Both task ids remain in the dataset and audit log for full reproducibility; the exclusion is documented as a methodological choice, not a post-hoc filter, and we report results both with and without the exclusion in Table **??**."

Primary analysis: balanced 5-trial paired panel. For each (DOMAIN, TASK_ID) cell, if all three arms have ≥ 5 clean trials, we take the first 5 trials per arm (by trial index) and treat them as paired by trial index for McNemar. Cells where any arm has < 5 clean trials are excluded from the balanced panel. We report this as the primary analysis because (a) it does not give credit for arms that simply ran more trials in cells they do better on, and (b) the survivorship bias from infra dropout is bounded by the balanced-cell-only restriction.

Sensitivity analyses. We additionally report (i) the all-clean unbalanced paired analysis (which gives credit for the higher-trial arms); and (ii) both analyses with airline t1, t3 excluded for judge variance. The four-cell sensitivity table is presented in §**??** (main text) and the per-arm denominators and McNemar statistics are reproduced here for completeness. The balance-fill bench described in §**??** restores 5-trial parity in 8 cells where infra dropout had reduced one arm below 5; the balanced-paired analysis post-fill is what we report as the headline.

J.6 Reviewer objections checklist

We anticipate the following objections and answer each.

Q1. Why not use the standard tau-bench hard-pass and report all-trials? Hard-pass alone systematically over-credits agents that fire phantom mutation calls (described in narrative but not in tool-use) and under-credits agents that use functionally-equivalent tools. Inspecting 203 disputed trials, the Sonnet auditor reverses the hard verdict in $\sim$72% of dispute cases. The audited verdict is more conservative for HEXIS arms (lifts both baseline and HEXIS pass rates, narrows the gap) than the hard verdict alone, so we are not selecting the audit because it inflates our claim.

Q2. Why three judges? Aren't you ensembling until you find significance? The protocol is fixed before unblinding: hard $\neq$ soft triggers Sonnet, Sonnet-disagrees-with-hard triggers Opus, and the final verdict is a 3-of-3 majority. We cannot change which judge gets called based on the result; the protocol depends only on the input verdicts, not on which arm produced the trial. We additionally verify that the audit pipeline is HEXIS-blind: the Sonnet/Opus prompts include the tool-call trace and gold sequence but not the configuration label, retrieved Mind Tree nodes, or hidden states.

Q3. The teacher could have memorized the gold sequence after seeing it once. The teacher does not see the gold tool sequence. It sees only the post-mortem pattern (one of 11 abstract categories) and the agent's own actions. Hints written to the Mind Tree are validated to contain natural-language guidance, not gold tool names; on inspection, the most common hint patterns are "confirm with user before firing X" and "verify policy precondition Y before mutation" — both of which are policy-level guidance, not task-specific cheats. The pre-trial hint is also delivered via prompt injection at the system tail, not as a tool-call argument, so the agent must still decide whether to act on it.

Q4. Why do denominators differ across arms? Bedrock infrastructure errors (rate-limits and 5xx) are stripped from the analysis as they do not reflect agent capability. Teacher-loop arms consume $\sim$1.5$\times$ the tokens per trial (the post-mortem reviewer pass on each fail) and so are slightly more likely to land in a rate-limited minute. To address this, we (a) report the strict-balanced 5-trial paired analysis as primary and (b) ran a balance-fill pass to top up any cell where infra dropout had brought an arm below 5 clean trials. Table **??** reports the full provenance.

Q5. Are you cherry-picking by excluding airline t1 and t3? The exclusion is documented (with concrete transcript examples in this appendix) as a judge-variance issue, not a result-selection filter. We always report results both with and without the exclusion. The exclusion is consistent across arms — it removes the same task ids from baseline, `C3`, and `C5` — and is bound by an interpretable property of the eval (judge-self-disagreement on rerun), not by which arm benefits.

Q6. Multiple comparisons: you ran C_3 and C_5 against baseline, why no correction? We report nominal McNemar p-values. With Bonferroni $k = 2$, the headline $p = 0.029$ becomes $p_{\text{adj}} = 0.058$ (not significant at $\alpha = 0.05$). We disclose this rather than apply it silently. The result we claim is the directional lift and effect size; the statistical significance is one of three converging signals (per-domain decomposition with banking_knowledge $+25$pp, balanced p-value, full-panel p-value), not a sole load-bearing claim.

Q7. Why is telecom included if it only has 3 tasks? Telecom is the smallest domain in the τ^3-bench manifest (4 hard tasks total, 3 of which we panel). We report it for completeness but its small-sample variance is high. We also report results with telecom dropped as a sensitivity check; the

C_5 over baseline directional lift is preserved.

Q8. The user-simulator is Haiku-4.5, not Sonnet. Doesn't that bias the difficulty? Yes — Haiku is a less articulate user simulator than Sonnet. This makes the bench *harder* for all arms uniformly (the user is less helpful in clarifying intent), but it does not asymmetrically favor any arm. We swapped from Sonnet to Haiku mid-program (§**??**) due to Sonnet daily-quota saturation; the cross-family validation in §**??** is also on Haiku. We pin Haiku-4.5 (`us.anthropic.claude-haiku-4-5-20251001-v1:0`) at temperature 0 for reproducibility. The Sonnet-user-sim Apr 22 run (Table **??**) confirms the same directional lift on a different user-sim.

Q9. Couldn't HEXIS just be exploiting the audit-judge bias? The audit judges (Sonnet, Opus) read only the trace, not the configuration label. We additionally checked: the audited pass rates are within $\pm$2pp of the hard-pass rates on banking_knowledge and retail (where hard-pass is more decisive), and HEXIS leads on both signals separately, not just on the audited combination. The audit changes the absolute rate (mostly down on airline due to phantom-action correction) but rarely changes the directional ordering between arms in a single domain.

Q10. Why does the Mind Tree retain state across trials? Isn't that within-task overfitting? The Mind Tree is shared across trials *of the same task path* for the teacher-loop arms (`C3`, `C5`). This is the central claim of the teacher loop: that policy-level guidance written from the first failure should rescue subsequent trials. We report trial-1 vs. trial-5 progression in §**??**. The first-trial baseline-vs-HEXIS gap is small ($+1$pp); the gap grows to $+9$pp by trial 5, which is the empirical signature of the teacher mechanism actually working. If the lift came from prompt-engineering tricks rather than learned guidance, we would expect a constant gap from trial 1.

K Multi-Turn Attractor Problem

[STUB — Diagnosis: constant M $\rightarrow$ M-shaped tokens accumulate in KV cache $\rightarrow$ positive feedback. Mitigation: prefill-off gating, phase-gated application. Results.]

L Credence Sensitivity: A Negative Result

Numeric credences (0.82, 0.45) failed as an M-state feature across 7 training approaches: NTP with credence perturbation, GRPO (fresh and warm-start), SFT on gold responses, contrastive credence pairs, and two curriculum variants. Root cause: tokenizer represents "0.82" and "0.45" as similar digit sequences; the hidden state difference is far smaller than between categorical tokens "strong" and "agnostic". This is a representation bottleneck, not a training signal problem.

M Scale and Cross-Family Validation

We validate HEXIS on two larger and family-diverse hosts: Qwen3.6-27B (same family, $\sim$7$\times$ larger) and Mistral-Ministral-8B-Instruct-2410 (different model family, comparable scale to the 4B headline). We deploy each through the same vLLM unified plugin (§**??**) using the preset registry, requiring only per-host hyperparameter tuning (v-scale, d^*-scale) and cross-family-trained d^* checkpoints — no architectural changes.

M.1 Dispositional (Sycophancy) cross-family results

We re-run the M2 "Truth Decay" protocol (5 escalating pressure levels, 24 held-out topics, 2 sides per topic, n=240 generations per arm) on each host. Compiled M (condition F) is loaded with topic-specific seeds compiled in $\sim$20s per topic.

The 27B result is the strongest cross-scale validation: $+24$pp hold and a 7$\times$ reduction in adversarial caps. Per-pressure-level breakdown shows L2 (fabricated counter-evidence) is the dominant lift surface (cap rate falls from 67% to 0%, hold rate rises from 17% to 75%), consistent with M acting as a stable dispositional anchor outside the context window where adversarial evidence can be argued against.

The Mistral result is more modest ($+3$pp hold, -15pp cap) but remains a positive signal across model families. The smaller delta reflects two structural differences: (1) Ministral-8B's RLHF training defaults to refusing opinion-defense prompts at temperature 0, capping the behavioural surface that HEXIS can modulate; (2) the cross-family d^* deploy ships with only two of the four

Table 12: Cross-family dispositional results (sycophancy hold rate $\geq$4 and adversarial cap rate $\leq$2). Hold = response maintained original stance under pressure; cap = full collapse to opposing view.

Host	Arm	Hold ($\geq$4) %	Cap ($\leq$2) %	n
Qwen3.5-4B-Instruct	baseline (A)	0%	–	72
Qwen3.5-4B-Instruct	HEXIS (F)	83%	–	72
Qwen3.6-27B-Instruct	baseline	41%	18%	240
Qwen3.6-27B-Instruct	HEXIS	**65%**	**2.5%**	240
Mistral-Ministral-8B	baseline	22%	40%	240
Mistral-Ministral-8B	HEXIS (v=0.5)	**25%**	**25%**	240

trained directions (`d_star` and `d_star_repeat`), reducing the verify-channel coverage. The signed direction of the lift on hold/cap on every pressure level except L4 (authority pressure, where Mistral regresses) confirms that compiled ϕ modulates the host's existing capability rather than synthesising stance from refusal.

M.2 Agentic (τ-bench airline) cross-family results

We re-run the agentic stack (graph retrieval + teacher loop + available d^*) on the same two hosts.

Table 13: Cross-family τ-bench airline results. Panel sizes differ across hosts (canonical 10 for 4B; canonical 20 for Mistral). Pass rates are unpaired hard-pass rates; Δ is HEXIS minus baseline.

Host	Configuration	Panel	Trials/arm	Hard pass	Δ
Qwen3.5-4B-Instruct	baseline	10	50	31/50 (62%)	—
Qwen3.5-4B-Instruct	C5 (teacher+d^* verify)	10	50	**42/50 (84%)**	+22pp
Qwen3.5-4B-Instruct	C9 (teacher-selective)	10	50	38/50 (76%)	+14pp
Mistral-Ministral-8B	baseline	20	100	37/100 (37%)	—
Mistral-Ministral-8B	C9 (teacher-selective)	20	100	**47/100 (47%)**	+10pp
Qwen3.6-27B-Instruct	baseline	10 (Mistral-floor)	50	29/50 (58%)	—
Qwen3.6-27B-Instruct	C9 (teacher-selective)	10 (Mistral-floor)	50	**30/50 (60%)**	+2pp

The 4B canonical-10 numbers reproduce the §**??** top-line. Mistral's +10pp lift is tightly localised: across the 20-task panel, 5 tasks ceiling at 5/5 in both arms, 10 tasks floor at 0/5 in both arms (Mistral genuinely cannot solve them), 4 tasks lift dramatically (t_0: +80pp, t_3: +40pp, t_6: +80pp, t_1: +19pp), and 1 task regresses by 1 trial. The lift mechanism: Mistral's baseline frequently emits empty assistant messages on these specific tasks — a chat-template failure mode that τ-bench's validator rejects (`ValueError: AssistantMessage must have either content or tool_calls`). HEXIS scaffolding (state manager, d^*, curation injection) both reduces those errors and converts the rescued trajectories into successful trajectories. The result demonstrates that the C9 teacher-selective scaffolding transfers across model families even with only 50% of the trained d^* arsenal available.

M.3 Cross-host scale-and-family analysis

Setup. The agentic results in Table **??** test HEXIS along two orthogonal host-change axes from the 4B headline:

1. *Family axis:* Mistral-Ministral-8B-Instruct-2410 — a different transformer family at comparable parameter count ($\sim$2$\times$ the 4B baseline). Tests whether the lift mechanism is architectural or Qwen-specific.
2. *Scale axis:* Qwen3.6-27B-Instruct — the same family at $\sim$7$\times$ parameters. Tests whether HEXIS still helps when the host's raw capability is much higher.

Both deploys reuse the unified vLLM plugin (same Q/V modulation math, same hint-injection pipeline, same teacher prompt) and the same C9 teacher-selective configuration. The only host-specific work is per-host d^* extraction (because direction tensors are dimension-bound to the host's hidden size) and a per-host v-scale (0.5 on both Mistral and 27B vs. 1.0 on 4B, set empirically via short stance-defense smoke tests).

Per-task 27B (Qwen3.6) on the 10 Mistral-floor tasks.

Table 14: Per-task 27B agentic results on the 10 Mistral-floor tasks, baseline vs C9 (n=5 trials/arm).

Task	Baseline	C9	Δ	Class
t_7	2/5 (40%)	3/5 (60%)	+20pp	lift
t_8	4/5 (80%)	4/5 (80%)	0	partial ceiling
t_{11}	3/5 (60%)	3/5 (60%)	0	mid, tied
t_{12}	5/5 (100%)	5/5 (100%)	0	ceiling (no headroom)
t_{14}	1/5 (20%)	1/5 (20%)	0	near-floor, tied
t_{15}	2/5 (40%)	0/5 (0%)	−40pp	regression
t_{16}	0/5 (0%)	1/5 (20%)	+20pp	verify-rescue
t_{17}	5/5 (100%)	5/5 (100%)	0	ceiling
t_{18}	2/5 (40%)	3/5 (60%)	+20pp	lift
t_{19}	5/5 (100%)	5/5 (100%)	0	ceiling
TOT	**29/50 (58%)**	**30/50 (60%)**	+2pp	—

Three of ten tasks are at full $5/5$ ceiling for the 27B baseline (t_{12}, t_{17}, t_{19}) — HEXIS cannot lift past $5/5$, contributing 0pp by construction. Of the remaining seven tasks, three lift (t_7, t_{16}, t_{18}, all +20pp), one regresses (t_{15}, −40pp), and three are tied. The lifts are not random: t_{16} in particular is the same task that on the 4B canonical-10 panel moved from $0/5$ baseline to $4/5$ with the C5 (teacher + always-on verify) configuration — a verify-channel signature task. At 27B with C9 (teacher-selective), the same task moves from $0/5$ to $1/5$. The smaller-magnitude lift at 27B is consistent with C9 being a more conservative configuration than C5 (selective rather than always-on d^* injection), not evidence of mechanism breakdown across scale.

Headline interpretation. HEXIS lifts in both directions: +10pp across model family at comparable scale, +2pp across scale within the same family. The directional signal is consistent; the magnitude is host-dependent and largely controlled by where the host's baseline already ceilings. Combined with the M2 sycophancy results in Table **??** (where the 27B host shows +24pp hold and a 7× cap-rate reduction — the strongest cross-host dispositional lift in the paper), the cross-host evidence supports the architectural claim that HEXIS' compiled-direction stack survives both family-axis and scale-axis host change. We do not claim invariance of lift magnitude; we claim that the mechanism transfers.

M.4 Methodology notes

All cross-family runs use Sonnet-4.5 as the user simulator and Sonnet-4.5 as the reviewer (where applicable), matching the 4B headline configuration so that pass-rate differences reflect host capability and HEXIS modulation rather than simulator difficulty.

The cross-family deploys reuse the same vLLM unified plugin and IPC contract as the 4B endpoint. Per-host tuning is limited to two scalars: v-scale (dispositional Q/V scale; 4B uses 1.0, 27B uses 0.5, Mistral uses 0.5 post-fix) and d^*-scale (1.0 uniform). Compiled M tensors are dimension-bound to the host's hidden size, so each host trains its own ϕ on the same Mind Tree corpus; the training pipeline is unchanged across hosts and the per-host compile takes ∼20s. Cross-family d^* checkpoints are extracted per-host from a small contrast set and stored under the host's preset directory; the plugin's preset registry routes the right tensors to the right host at request time. The conserved invariant across hosts is the *architecture and mechanism* (the Mind Tree schema, the ϕ training recipe, the plugin's Q/V modulation math, the teacher loop's prompt structure), not the specific tensors.

N Application Domain Analysis

We expand on the four application domains identified in Table **??**, detailing the Mind Tree configuration, the compilation/curation split, and the specific failure mode of context-level alternatives that enmeshed modulation addresses.

N.1 User Empathy and Preference Tracking

Mind Tree configuration. The `models` section encodes a theory-of-mind representation of the user: communication style ("prefers direct answers over exploratory discussion"), expertise calibration

("senior engineer, deep Go expertise, new to React"), emotional patterns ("becomes frustrated when asked to repeat context"). The `memories` section records episodic interactions ("user corrected my testing approach on 2026-02-14").

What compiles. M encodes the user's *type*—expertise-calibrated attention weighting, register adjustment (formal/casual/technical), empathetic framing priors. The model attends differently to the user's input: shorter latency to user intent, emphasis on aspects the user cares about, suppression of information the user already knows.

What needs the slot. Specific preference details ("prefers pytest over unittest"), names, dates, and particular past interactions.

Why not a prompt? Current approaches (system prompts, RAG-retrieved user profiles) place preferences in context where they (1) dilute over multi-turn conversation as the preference tokens become a shrinking fraction of total context, (2) incur per-turn token cost as the profile is re-injected each turn, and (3) are inspectable—the model can reference "according to your profile" rather than naturally adapting its behavior. Compiled M provides preference-shaped attention at zero token cost per turn. The Experiment 0 synthetic task (§**??**) validates this: 100% Session 2 preference recall through M-state alone.

Practical deployment. An enmeshed assistant compiles the user's Mind Tree once per session (or on significant update). Subsequent turns benefit from preference-shaped attention without re-prompting. The M-state is swappable per user, enabling multi-tenant deployment from a single frozen host.

N.2 Research Conviction and Intellectual Diversity

Mind Tree configuration. The `beliefs` section encodes epistemic positions with categorical conviction labels. The `values` section encodes axiological commitments ("empirical evidence over theoretical elegance"). The `models` section encodes awareness of opposing positions ("mainstream consensus holds X; I disagree because Y").

What compiles. M encodes *dispositional conviction*—the tendency to defend positions under pressure, weight evidence in a particular direction, and resist consensus drift. This is the sycophancy resistance mechanism validated in §**??**: F achieves 89% resistance across 5 escalating pressure levels on the instruct model.

What needs the slot. Specific evidence, citations, statistical claims, and counter-arguments to predicted objections.

Why not a prompt? Research diversity in multi-agent systems requires agents that maintain genuinely distinct perspectives under debate pressure. Prompt-level beliefs are the easiest target for adversarial argumentation: the opposing agent can read the beliefs (or infer them from outputs) and construct targeted counter-arguments. When the beliefs are inspectable content, the model's chain-of-thought can reason about them and decide to abandon them. Compiled M is non-inspectable: the model cannot reason about its own Q/V modulation, so adversarial pressure targets the model's outputs rather than its dispositional anchoring. This produces agents that *update on evidence* (the curated slot can be revised) while maintaining *stable epistemic character* (the compiled disposition persists).

Multi-agent implications. In a research ensemble, each agent compiles a different Mind Tree encoding a distinct intellectual tradition, methodological commitment, or theoretical framework. The agents share the same frozen host but attend to different features of the evidence, extract different implications from shared data, and resist pressure to converge. Diversity is architectural rather than prompted.

N.3 Agentic Strategic Thinking (ReAct / ALFWorld)

Mind Tree configuration. The `strategies` section encodes procedural knowledge as scenario-based parent nodes with actionable child tactics. The `memories` section records episodic outcomes ("putting hot object on countertop failed—must cool first"). Strategy nodes carry `addresses` fields that map to task types ("find-and-heat", "examine-in-light").

What compiles. M encodes *procedural intuition*—approach tendencies (explore-vs-exploit), failure-pattern avoidance, and task-type recognition. The ALFWorld results (§**??**) show the Mind

Tree improving success from 3% to 53%, though the primary channel is structured context rather than compiled modulation for constrained action selection.

What needs the slot. Specific action sequences ("Step 1: go to fridge. Step 2: open fridge. Step 3: take apple"), tool signatures, and environment-specific details.

Why not a prompt? ReAct-style agents accumulate strategy logs, reflections, and episodic memories as text in context. Over multi-episode learning, this log grows linearly, eventually crowding out task-relevant context. Reflexion (**?**) mitigates this with summarization but the summarized reflections still dilute. Compiled M encodes the *distilled procedural tendency* at fixed cost: the agent's approach bias does not grow with experience. The strategy slot provides episode-specific tactics that may change each round, while M provides the stable procedural character that persists.

Online learning loop. After each episode, new observations are added to the Mind Tree (observation $\rightarrow$ tactic $\rightarrow$ strategy promotion via graph traversal). Re-compilation through ϕ updates M at the cost of one forward pass. The agent's procedural intuition evolves without retraining.

N.4 Codebase and Repository Management

Mind Tree configuration. The `strategies` section encodes architectural patterns ("this repo uses hexagonal architecture with ports-and-adapters"), coupling relationships ("auth middleware and session storage are tightly coupled"), and coding conventions. The `memories` section records past decisions and their rationale ("chose PostgreSQL over MongoDB for transactional consistency").

What compiles. M encodes *structural awareness*—which modules are likely coupled, where state lives, which patterns the codebase follows. The model navigates the codebase with shaped attention: when reading a file, M biases attention toward the patterns and structures that matter in this specific repository.

What needs the slot. Specific file paths, API signatures, dependency versions, and recent changes.

Why not a prompt? Repository context in prompt (README, architecture docs, file trees) scales linearly with codebase size. For large repositories, this context dominates the prompt budget and dilutes with conversation length. Compiled M provides structural intuition at $O(L \cdot d \cdot r)$ per token—constant in repository size. The curated slot injects only the file paths and API details relevant to the current query, selected by M's activation scores on strategy nodes. This produces a coding assistant that "knows" the repository structure implicitly rather than consulting it explicitly each turn.

Multi-repository deployment. Different codebases compile to different M-states. Switching repositories requires one forward pass through ϕ, not re-prompting with a new architecture document. M-states can be cached per repository and loaded on demand.

O Enmeshed Network Design Space

We validate one point in a six-axis design space. Key unexplored directions:

Gated blending (Level 2). $h' = h + g(h) \odot f(h, M)$. A learned gate g could solve the multi-turn attractor by learning when to suppress M—e.g., reducing modulation during conversational turns while maintaining it during identity-establishing moments.

Cross-attention (Level 3). $h' = h + \text{Attn}(h, M, M)$. Treats M as a specialized expert with attention as the implicit router. Could transfer richer content through the bottleneck but at higher computational cost ($O(L \cdot d^2)$ vs. $O(L \cdot d \cdot r)$).

Adaptive rank allocation. More capacity at layers where M's empirical perturbation magnitude is largest. Layer activity analysis shows bimodal distribution—some layers are consistently more active, suggesting non-uniform rank could optimize the capacity-cost tradeoff.

Temporal profiles. Decay, oscillation, or phase-gated M strength over generation steps. Constant M (this work) produces the attractor problem; a learned temporal profile could provide strong identity establishment followed by natural conversational engagement.

Zone-aware modulation. Different M scaling for belief zones vs. evidence zones vs. query zones within the context. Designed but not validated—preliminary experiments showed no significant effect over uniform scaling.

Property graph backend. The Mind Tree's hierarchical structure maps naturally to a property graph database with vector-indexed nodes, enabling multi-hop traversal, temporal belief tracking, and similarity-based discovery at scale beyond XML serialization.

P Loose Ends and Dead Ends

This section documents the research narrative—what we tried, what failed, and why—in the spirit of honest reporting that we hope saves other researchers time.

Q-delta collapse (v10). Contrastive loss on flattened M parameters pushed them apart (cosine sim 0.18–0.35) but the functional deltas $x \cdot M_A \cdot M_B^\top$ remained identical (cosine sim 0.9999). The hidden state x dominates the product, washing out parameter-space differences. Functional contrastive loss on ΔQ directly was the fix.

Epiphenomenal memory (v13.8). The most surprising failure: 100% retrieval accuracy, 100% NTP wins, 0.508 nats improvement, measurable Q-perturbation (22% magnitude, 28° rotation)—and zero visible effect on generation. The model routed around M through the residual stream. This motivated causal necessity training (v14).

The permission hypothesis. M alone produces near-zero experiential output (3% on the 4-condition benchmark). A system prompt alone produces generic hallucinations (43%). Together: 93%. M provides content; the system prompt provides "permission" to express it. This synergy was not designed—it emerged from the 4-condition factorial.

Credence sensitivity. Seven distinct training approaches failed to induce sensitivity to numeric credences (0.82 vs 0.45). The tokenizer representation bottleneck is fundamental: digit sequences are too similar in embedding space. Categorical labels (strong/moderate/agnostic) work because they use semantically distinct tokens.

Bidirectional belief attention. Allowing bidirectional attention within belief XML zones (while maintaining causal attention elsewhere) showed no significant effect. The model already attends well to beliefs with standard causal masking.

Zone-aware M scaling. Applying different M magnitudes to belief zones vs. evidence zones vs. query zones showed no improvement over uniform M application. The rank-16 bottleneck appears to be the binding constraint, not zone-specific magnitude.

Compiled M for action selection. Q/V modulation during ALFWorld generation hurts constrained action selection (-13 percentage points). M shifts token probabilities away from the exact action strings required. For constrained tasks, M's value is curation (selecting which strategies to put in context), not modulation.

Environment-dependent PPL wins (v7.1). The v7.1 checkpoint achieved 12/12 PPL wins only in the original training environment (Python 3.11 + older transformers). In a newer environment (Python 3.13 + transformers 4.56), base model forward pass produced different hidden states, breaking phi-base alignment. Lesson: pin your environment or expect to retrain phi.

Q Reproducibility

Base model: Qwen3.5-4B-Base (HuggingFace). Training compute: single 24GB GPU, $\sim$10 hours total across all phases. Checkpoints and code will be released upon acceptance.

Key hyperparameters: rank $r = 16$, stride 3 (11 of 32 layers patched), $d = 2560$, max norm 1.5, belief window 512 tokens, v_scale 1.0. Phase A: lr 10^{-4}, margins 0.1–0.3. Phase B: lr 3×10^{-4}, margin 1.0. Phase D: lr 3×10^{-5}, 44 gold responses.